
The Convergence Cost of Reasoning: Why Per-Token Early Exit Underperforms

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Abstract

1 Per-token early exit collapses to near-zero question accuracy (Q_{Acc}) on reasoning
2 models, where compound error across thousand-token traces defeats all standard
3 CALM criteria. Via thinking mode ablation on one architecture (Qwen3-8B), we
4 decompose two failure mechanisms: oscillatory dynamics (distillation-specific;
5 absent under native reasoning) and representation immaturity (reasoning-general).
6 Tuned lens analysis reveals a +45.6 pp oracle gap, identifying projection quality as
7 the primary bottleneck. Under nested cross-validation (outer fold trains tuned lens,
8 inner fold trains gate), a learned gate recovers 48–63% Q_{Acc} at ~5–10% savings
9 (exit-layer-grid-dependent)—modest but nonzero versus CALM’s 0%—across four
10 models spanning three architecture families (Qwen, Llama, Phi) at 8B–32B scale;
11 the 7B distilled variant and non-reasoning control show chance-level gate loss.
12 Standard CALM criteria fail at all scales tested (8B–32B); QwQ-32B collapses to
13 2.4% Q_{Acc} , yet V2a recovers 61.0% at ~5% savings. Cross-benchmark replication
14 on GSM8K confirms directional consistency (Appendix Z).

15 1 Introduction

16 Per-token early exit—one of the most promising inference optimizations for long reasoning traces—
17 collapses to 0% question accuracy on current reasoning models. Models such as DeepSeek-
18 R1 [DeepSeek-AI, 2025] and QwQ [Qwen Team, 2024] routinely exceed 3,000 tokens per question,
19 and compound error across such sequences is well known [Bengio et al., 2015, Ranzato et al., 2016]:
20 even 99.2% per-token accuracy yields 0% question-level accuracy over ~3,000 tokens. However, this
21 observation alone does not explain *why* compound error is so severe in reasoning traces, nor does it
22 suggest *how* to mitigate it. Reasoning traces are not homogeneous: they interleave numerical com-
23 putation, logical connectives, and formatting scaffolding—token roles with fundamentally different
24 convergence dynamics that prior work has not fully characterized. At the sequence lengths typical of
25 translation or summarization (50–100 tokens), 99.2% per-token accuracy yields ~45–67% all-correct
26 sequence accuracy—degraded but tolerable under soft metrics; at 3,000 tokens, the same rate yields
27 effectively 0%—a qualitative rather than quantitative shift.

28 Comparing 2 architectures $\times$ 2 training conditions, we decompose two failure mechanisms with
29 architecture-dependent reversal: oscillatory dynamics in numerical tokens produce false convergence
30 signals, while reasoning keywords exhibit representation immaturity despite appearing confident—
31 two independent mechanisms that undermine any single exit threshold (pseudo- $R^2 \leq 0.10$, $p \approx 0$).

32 We term the resulting shortfall the *Reasoning Early Exit Gap*—the deficit in question-level accuracy
33 caused by per-token layer exit on reasoning traces, as quantified by the tuned lens oracle gap (+45.6 pp
34 over raw projection; §5.3). A learned gate modestly recovers from this gap (48–63% Q_{Acc} at ~5–10%
35 savings (exit-layer-grid-dependent) under nested cross-validation across four models spanning three
36 architecture families; §5.2), but the deficit between these modest savings and the 58% oracle ceiling

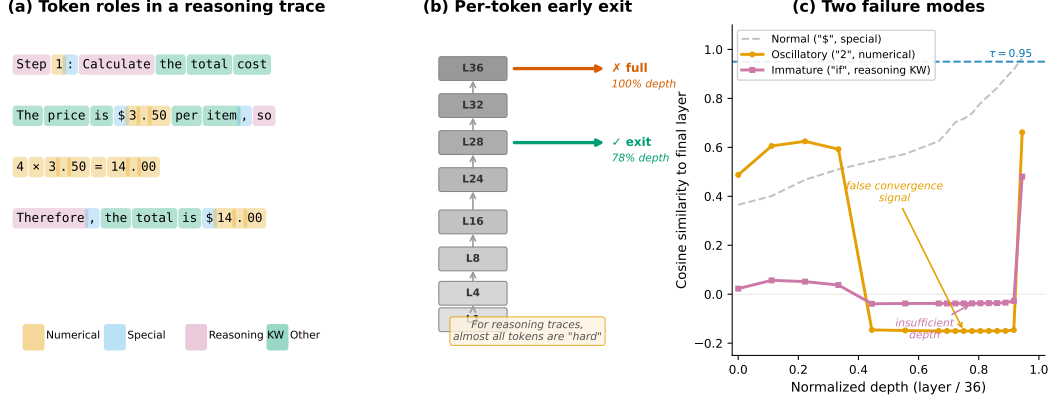


Figure 1: Overview of our analysis framework. (a) Token roles in reasoning traces: four categories with distinct convergence properties. (b) Per-token early exit (schematic): each token can potentially exit at an intermediate layer, but reasoning tokens resist early convergence. (c) Two failure modes identified: oscillatory dynamics (primarily in distilled models) co-occur with false convergence signals, while representation immaturity (across all reasoning models) limits tokens from reaching exit thresholds at sub-final layers. Real per-token cosine-similarity trajectories from R1-Qwen3-8B on GSM8K (selected to illustrate typical failure modes; see Figure 2 right for population means).

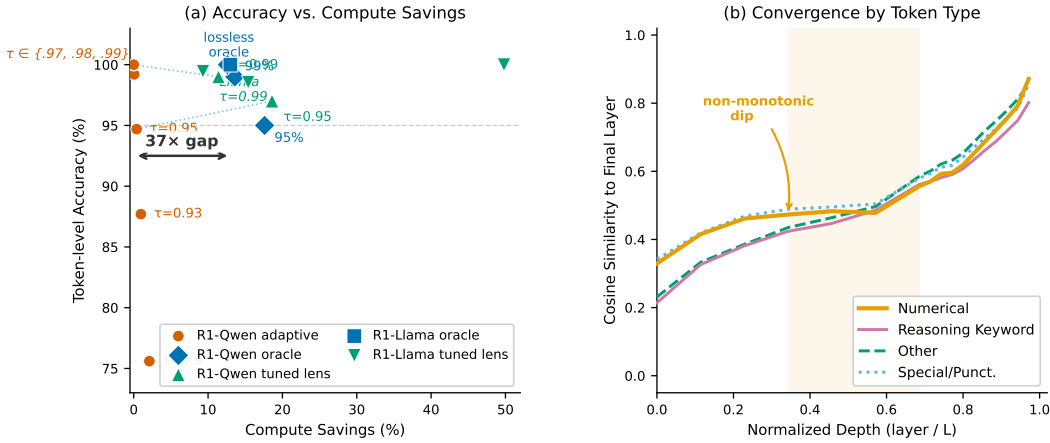


Figure 2: **Left:** Question-level accuracy vs. compute savings for cosine-similarity exit (near-zero savings) vs. oracle (12.6% lossless, 17.6% at 95% token accuracy). **Right:** Mean cosine similarity to final layer by token role in R1-Qwen3-8B, showing non-monotonic dip in numerical tokens. Data: 500 questions, 1.49M tokens.

underscores the depth of the challenge. These patterns extend to code reasoning (HumanEval) and 14B models.

Contributions.

- Dual failure pattern:** we observe, *on one architecture* (Qwen3-8B, the only one with a native thinking mode comparator), oscillatory dynamics associated with distillation and representation immaturity (reasoning-general, persisting without distillation), and show that both are separable via thinking mode ablation (§4.4).
- Projection bottleneck:** tuned lens reveals projection quality as the primary bottleneck (+45.6pp on GSM8K, +46.7pp on MATH-500). A learned gate recovers 48–63% Q_{Acc} at ~5–10% savings (nested CV; exit-layer-grid-dependent) across four models spanning three architecture families at 8B–32B scale (§5.2); compound error remains the binding deployment constraint. *Scope:* 8B–32B (full V2a analysis including QwQ-32B). Reverses at 1.5B (§7).

2 Related Work

Early exit for Transformers. CALM [Schuster et al., 2022] proposed confidence-based adaptive exit; we show standard CALM measures yield at most 20% Q_{Acc} at $\leq 0.03\%$ savings on reasoning traces (§5). LayerSkip [Elhoushi et al., 2024] combines early exit with self-speculative decoding; subsequent work extends exit along multiple axes [Graves, 2016, Del Corro et al., 2023, Raposo et al., 2024, Fan et al., 2024, Jiang et al., 2024, Xu et al., 2025, Yang et al., 2025b, Li et al., 2024, Bae et al., 2025, Noh et al., 2026]. GateSkip [Laitenberger et al., 2026] and Wei et al. [2026] independently observe reasoning-task degradation; FlexiDepth [Luo et al., 2025] notes token-role-dependent computation demands; TIDE [Jaber and Jaber, 2026] applies learned routers to reasoning models.¹ Song et al. [2025] document deeper-layer reliance; our contribution is a quantitative decomposition that documents architecture-dependent reversal of two failure mechanisms.

Concurrent work TIDE [Jaber and Jaber, 2026] attaches learned routers for per-token layer exit on reasoning models, reporting 98–99% task accuracy retention with 6–8% throughput gains on R1-Distill-8B. However, TIDE evaluates task-level accuracy (final-answer correctness) rather than per-token exit fidelity (Q_{Acc}): task accuracy tolerates intermediate token errors that do not affect the final answer, and TIDE tests only a single distilled architecture. Our analysis complements TIDE by providing the diagnostic foundation—identifying *why* per-token exit degrades reasoning accuracy and *where* the representation bottleneck lies.

Sequence-level reasoning efficiency. Orthogonal to per-token *layer* exit, sequence-level methods decide when to terminate chain-of-thought generation [Yang et al., 2025a, Jiang et al., 2025, Nagle et al., 2026, Sui et al., 2025, Li et al., 2025] or adapt reasoning depth to difficulty [Liu et al., 2025]; S-GRPO [Dai et al., 2025] and Step-GRPO [Chen et al., 2026a] use reinforcement learning to learn early exit policies for reasoning chains. These are complementary: they decide *how many tokens* to generate, while we analyze *at which layer* each token can exit.

Reasoning-capable LLMs. Chain-of-thought prompting [Wei et al., 2022] has been scaled through specialized training [OpenAI, 2024, DeepSeek-AI, 2025, Qwen Team, 2024], producing traces 10–100 $\times$ longer than direct answers. Qian et al. [2025] identify MI-peak “thinking tokens” that drive reasoning, and Bogdan et al. [2025] show that reasoning steps contribute unequally to final answers; Lin et al. [2024] identify critical tokens via rollout sampling. Our analysis connects token roles to layer-exit feasibility.

Layer-wise representation analysis. The logit lens [nostalgebraist, 2020] projects intermediate hidden states through the LM head; the tuned lens [Belrose et al., 2024] adds per-layer affine transforms. Song et al. [2025] show reasoning relies on deeper layers; Nepal et al. [2025] find critical math-reasoning layers forged during pre-training. Layer redundancy analyses [Men et al., 2024, Yang et al., 2024] show many layers can be pruned on standard benchmarks, yet reasoning tokens resist such compression. Concurrently, DTR [Chen et al., 2026b] and DRTC [Chang, 2026] address per-token importance and critical trace segments; our work is complementary (matched-model comparison, dual failure mode decomposition).

3 Methodology

3.1 Experimental Design: 2 $\times$ 2 Matched Comparison

To separately characterize the associations of model architecture and reasoning-specific training, we employ a 2 $\times$ 2 matched comparison crossing two architectures with two training conditions at the 8B scale (Table 1; scope discussed in §7).

The reasoning models are distilled from DeepSeek-R1 (671B; R1-Qwen uses the 0528 release), producing extended traces ($\sim 3,000$ tokens per question on GSM8K). The non-reasoning models are instruction-tuned counterparts controlling for output quality. Our design is observational: both distilled reasoning models share a single teacher (DeepSeek-R1 671B), so the 2 $\times$ 2 design cannot distinguish reasoning-training effects from teacher-model effects. Our “training condition” factor

¹TIDE and Probe and Skip [Wu et al., 2026] are concurrent work.

Table 1: 2×2 matched comparison. Each cell is an 8B model; cross-architecture comparisons use relative depth $d = L/L_{\text{total}}$.

	Qwen (36 layers)	Llama (32 layers)
Reasoning	DeepSeek-R1-0528-Qwen3-8B (R1-Qwen) ^a	DeepSeek-R1-Distill-Llama-8B (R1-Llama) ^b
Non-reasoning	Qwen3-8B [Qwen Team, 2025] (Qwen3) ^c	Llama-3.1-8B-Instruct (Llama) ^d

^adeepseek-ai/DeepSeek-R1-0528-Qwen3-8B ^bdeepseek-ai/DeepSeek-R1-Distill-Llama-8B
^cQwen/Qwen3-8B ^dmeta-llama/Llama-3.1-8B-Instruct

captures the *net* association; the native thinking mode experiment (§4.4) confirms the deficit without distillation.

Data. We evaluate on GSM8K [Cobbe et al., 2021], MATH-500 [Hendrycks et al., 2021], HumanEval [Chen et al., 2021], and AIME 2024 (Appendix R). We profiled 100–500 questions per model; R1-Llama serves as a cross-architecture check with wider bootstrap CIs ($\pm 1.6\text{pp}$ vs. $\pm 0.3\text{pp}$; Appendix B). Phi-4-reasoning uses a 16K-token budget (truncation artifact $\leq 3\text{pp}$; sample details in Appendix V, N).

3.2 Token Role Taxonomy

We classify each token in a reasoning trace into four mutually exclusive categories via deterministic regex rules applied to the decoded token string:

- **Numerical:** digits, decimal points, arithmetic operators, and numeric expressions (e.g., 3.14, +, =). These carry the computational payload.
- **Special:** punctuation, brackets, newlines, and structural markers (e.g., \n, <think>, ,). These provide formatting scaffolding.
- **Reasoning keyword:** logical connectives and reasoning markers (e.g., “therefore”, “because”, “hence”, “so”). These signal reasoning transitions.
- **Other:** all remaining tokens, predominantly natural language content.

Token role is a statistically significant but modest correlate of exit outcomes (pseudo- $R^2 \leq 0.10$, $p \approx 0$; Appendix Y); critically, effect directions reverse across architectures (numerical OR=6.76 in R1-Qwen vs. 0.20 in Phi-4), so no universal per-type exit policy can serve both—motivating the learned gate (§5.2). The taxonomy is a decomposition tool for exposing architecture-dependent failure modes (§4), not a predictive framework.

3.3 Metrics

Non-monotonic convergence rate. For each token t , let $s_\ell = \cos(\mathbf{h}_\ell^t, \mathbf{h}_{\text{final}}^t)$ denote the cosine similarity between the hidden state at layer ℓ and the final layer. We maintain a running maximum $M_\ell = \max_{j \leq \ell} s_j$ and flag token t as *non-monotonic* if $\exists \ell : M_{\ell-1} - s_\ell > \theta$ (default $\theta = 0.03$; sensitivity analysis in Appendix A)—i.e., the similarity at any layer falls below its historical peak by more than θ . Such tokens violate the monotonic convergence assumption underlying all confidence-based exit methods. All confidence intervals are computed via question-level bootstrap resampling (1,000 iterations).

Exit accuracy. At each checkpoint layer L , we project the intermediate hidden state through the final LM head: $\hat{y}_L = \arg \max(\mathbf{W}_{\text{lm}} \cdot \mathbf{h}_L)$. *Token-level exit accuracy* is the fraction of tokens where $\hat{y}_L = \hat{y}_{\text{final}}$. *Question-level exit accuracy* (hereafter Q_{Acc}) requires *all* tokens in a question’s reasoning trace to match. A segment-level analysis (Appendix S) confirms this is not merely a strict criterion: even evaluating only the final boxed answer, question accuracy drops from 84.3% (full depth) to $< 2\%$ at L32; compound error across multi-token sequences is the primary bottleneck.

Proxy validation. We compute Spearman ρ between per-token convergence speed (earliest layer achieving $\cos \geq 0.95$) and binary exit correctness at each layer. An effective proxy requires strong positive ρ . With 96 tests (4 models $\times$ 6 layers $\times$ 4 roles), we apply Benjamini–Hochberg FDR correction at $\alpha = 0.05$; all reported correlations remain significant at $q < 0.01$.

Table 2: Non-monotonicity rates ($\theta = 0.03$) across the 2×2 design, Phi-4-reasoning 14B cross-architecture validation ($n=180$, 2.8M tokens), and native thinking mode ablation. Token-level rates; ANOVA uses question-level means (4×100 balanced, excludes thinking mode and Phi-4).

Model	Training	Overall	Numerical	Special	Reas. KW	Other
R1-Qwen3-8B	reasoning	20.8% *	42.9%	27.4%	4.2%	11.8%
R1-Llama	reasoning	19.3%	11.7%	34.1%	11.7%	11.5%
Qwen3-8B	non-reas.	11.6%	15.3%	18.9%	7.8%	8.5%
Llama-3.1-8B	non-reas.	8.5%	1.6%	8.2%	11.8%	9.6%
<i>Cross-architecture validation (14B, 40 layers)</i>						
Phi-4-reasoning	reasoning	25.5%	5.6%	43.6%	3.0%	22.3%
Phi-4 (base)	non-reas.	11.6%	8.1%	7.9%	1.2%	14.0%
Qwen3-8B (thinking)	native reas.	9.1%	16.0%	8.3%	11.5%	7.3%

*Pooled rate across GSM8K ($n=750$) and MATH-500 ($n=364$).

Relative depth. Cross-architecture comparisons use normalized depth $d = L/L_{\text{total}}$ (e.g., L28 = $d_{0.88}$ for 32-layer, $d_{0.78}$ for 36-layer models).

4 Findings

4.1 Non-Monotonic Convergence

A token’s cosine similarity trajectory $\cos(\mathbf{h}_L, \mathbf{h}_{\text{final}})$ is *non-monotonic* if it dips by $\geq \theta$ before recovering—indicating that the representation moves *away* from its final state at intermediate layers. Non-monotonicity is one observable symptom associated with reasoning training; §4.3 shows it is associated with—but does not fully account for—the exit accuracy deficit, as a second, distinct pattern is also observed.

At the 8B–14B scale, reasoning training is consistently associated with higher non-monotonicity (Table 2), but with architecture-dependent magnitude: Llama more than doubles (8.5%→19.3%), while Qwen increases modestly (11.6%→20.8%). Within our design sharing a common teacher, training condition is the stronger correlate of non-monotonicity (Appendix C); this association captures the net effect of reasoning training, distillation, and teacher-specific factors—it is not a causal estimate.

Reasoning-trained models exhibit $\sim 2\times$ the non-monotonicity of non-reasoning counterparts (Figure 3, left; stable across three disjoint subsets, max $\Delta < 1\text{pp}$; Appendix A). The pattern holds across all $\theta \in \{0.01\text{--}0.10\}$ (Appendix A). Non-monotonicity is sharply stratified by token role but the dominant role varies: in R1-Qwen, **numerical tokens reach 42.9%** vs. reasoning keywords 4.2% (**10.2 $\times$ gap**); in Phi-4-reasoning, **special tokens lead at 43.6%**. Per-type rankings are model-specific, precluding universal exit policies; robust after controlling for chain length (Appendix E, W, A). Effect directions **reverse across architectures**: numerical OR=6.76 in R1-Qwen but 0.20 in Phi-4, confirming training-paradigm-dependent mechanisms (logistic regression details in Appendix Y).

Cross-architecture and scale validation. Phi-4-reasoning (14B) confirms the pattern: non-monotonicity more than doubles versus its base (25.5% vs. 11.6%; Table 2), though the dominant token role differs (special vs. numerical). At 1.5B scale, this pattern reverses (Appendix V): both the non-monotonicity gap and exit accuracy deficit invert, constraining generalization to 8B–14B.

4.2 Exit Accuracy Deficit

Reasoning training is associated with lower token-level exit accuracy across both architectures studied at the 8B–14B scale.

Reasoning-trained models consistently show lower exit accuracy (Table 3): R1-Qwen (32.4% [32.1, 32.7]) trails Qwen3-8B by 15.3pp, R1-Llama trails Llama (74.3% [72.8, 76.0]) by 6.4pp (non-overlapping 95% bootstrap CIs). The magnitude is architecture-dependent—Qwen’s larger deficit is

Table 3: Token-level exit accuracy (question-level mean) at L28 across the 2×2 design. L28 = 78% depth for Qwen, 88% for Llama; Δ controls for depth within architecture.

	Qwen (L28 = $d_{0.78}$)	Llama (L28 = $d_{0.88}$)
Reasoning	32.4%	67.9%
Non-reasoning	47.7%	74.3%
Δ (reasoning – non-reasoning)	–15.3pp	–6.4pp

consistent with L28 at 78% vs. 88% relative depth. Token role significantly predicts exit correctness at every layer in R1-Qwen3-8B ($p \approx 0$), with the effect attenuating at deeper layers (Appendix Y).

14B validation and per-type patterns. Phi-4-reasoning (14B) confirms the deficit (–8.3pp at L8; Appendix V), with a phase transition at ~ 40 –50% depth mirroring the 8B pattern. Reasoning training compresses per-type exit accuracy from a 26.3pp range to a 5.2pp band; reasoning keywords show the largest drop (–28.8pp) yet the lowest non-monotonic rate (4.2%)—smooth but insufficiently deep convergence, the second failure mode of §4.3 (details in Appendix O, O).

Tuned lens (per-layer affine transforms; Belrose et al. 2024; 5-fold nested CV: 400 train / 100 test per fold) confirms the deficit is genuine, not an artifact of raw projection. Tuned lens raises absolute exit accuracy by +35–54pp (e.g., L24: 11.7%→65.7%), identifying raw projection quality as the primary bottleneck (see §5.3 for distribution analysis). Despite this improvement, the reasoning–non-reasoning gap *persists* at every layer (4–6pp at L24–L32; Appendix T)—the deficit is a property of reasoning representations themselves.

4.3 Proxy Failure and Dual Failure Pattern

The standard cosine-similarity proxy fails to predict exit correctness in reasoning models: for R1-Qwen, $|\rho| < 0.15$ at every layer (Figure 3, right), confirming that cosine similarity is uninformative about exit correctness. R1-Llama’s overall $\rho = +0.32$ at L24 masks a per-type inversion—reasoning keywords show $\rho = -0.51$. An entropy-based alternative is complementarily blind: reasoning keywords have low entropy yet the worst exit accuracy.

Two independent failure patterns. Two empirically distinct failure patterns characterize the exit accuracy deficit, each affecting different token populations: **(1) Oscillatory dynamics** (distilled models): numerical and special tokens exhibit non-monotonic trajectories—appearing converged then diverging—more pronounced in distillation-trained models (§4.1) but largely absent in native reasoning (§4.4); **(2) Representation immaturity** (all reasoning models): reasoning keyword tokens converge monotonically but fail to reach sufficient accuracy at sub-final layers (L32: 55.2%), with low entropy falsely suggesting confidence. No single threshold resolves both: lowering τ triggers false exits from oscillating tokens, while raising τ blocks insufficiently-converged tokens indefinitely (Appendix D). The independence of these failure patterns is confirmed by the native reasoning experiment (§4.4).

4.4 Native Thinking Mode Ablation

A native thinking mode ablation (Qwen3-8B only; Llama-3.1-8B lacks a native thinking mode, $n=499$) decomposes the two failure modes. Oscillatory dynamics are distillation-specific: thinking NM 9.1% is *below* non-reasoning 11.6%, compared to R1’s 20.8% (Table 2)—the elevated oscillation is absent without distillation. Representation immaturity persists without distillation (L28 exit accuracy 34.7% $\approx$ R1’s 32.4%, both $\ll$ non-reasoning 47.7%; Appendix U), confirming it as reasoning-general. The ANOVA η^2 (§4.1) cannot disambiguate training from teacher identity; this ablation provides a partial control on one architecture (Qwen only; cross-architecture generalization requires future work).

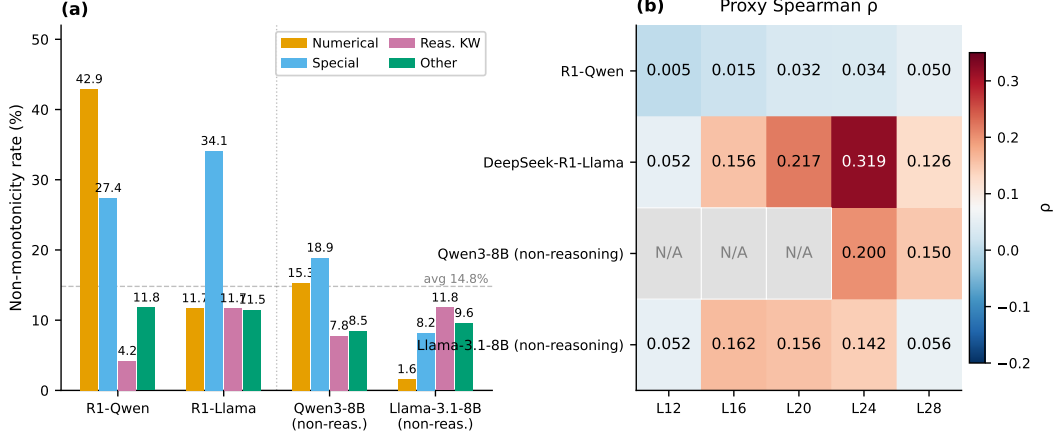


Figure 3: **Left:** Non-monotonic convergence rates by token role across the 2×2 design (R1-Qwen3-8B, R1-Llama-8B, Qwen3-8B, Llama-3.1-8B; 500 questions each except Qwen3 $n=499$). **Right:** Cosine-similarity proxy Spearman ρ heatmap. R1-Llama’s moderate $\rho=0.32$ masks a per-type inversion ($\rho=-0.51$ for reasoning keywords).

Table 4: Adaptive exit under raw CALM criteria and tuned lens. Softmax criteria: $n=50$; cosine similarity and tuned lens: R1-Qwen $n=500$, R1-Llama $n=500$.

Projection	τ	R1-Qwen (500q)			R1-Llama (500q)		
		Tok Acc	Q_{Acc}	Savings	Tok Acc	Q_{Acc}	Savings
Raw (softmax resp.) [†]	0.99	86.9%	0%	8.1%	—	—	—
Raw (softmax gap) [†]	0.99	88.9%	0%	7.2%	—	—	—
Raw (cos-sim)**	0.95	94.7%	0%	0.39%	—	—	—
Raw (cos-sim)**	0.97	99.2%	0%	0.054%	—	—	—
Raw (per-type)	mixed [†]	95%	0%	0.34%	—	—	—
Tuned lens	0.95	97.0%	7.2%	18.6%	98.6%	20.2%	15.4%
Tuned lens	0.99	99.0%	29.4%	11.4%	99.5%	52.0%	9.3%

[†]“mixed”: per-type optimized thresholds. R1-Qwen: 36 layers, 9 exit points; R1-Llama: 32 layers, 6 exit points.

[†]CALM criteria ($n=50$); Q_{Acc} 0% at all $\tau \in \{0.5, 0.7, 0.8, 0.9, 0.95, 0.99, 0.999, 0.9999\}$.

**Exit-to-final-layer cosine similarity ($n=500$), adapted from CALM [Schuster et al., 2022]; Q_{Acc} 0% at $\tau \geq 0.95$. At $\tau=0.9$ ($n=50$): 20% Q_{Acc} , 0.03% savings.

R1-Llama “—”: raw CALM criteria evaluated on R1-Qwen only ($n=50$); softmax-based 0% Q_{Acc} on R1-Qwen ($n=50$); not evaluated on R1-Llama (R1-Llama: 52.0% Q_{Acc} vs. 0% raw).

208 5 End-to-End Early Exit Analysis

209 We now test progressively stronger exit strategies on R1-Qwen3-8B (500 GSM8K questions), from
 210 standard confidence methods (§5.1) to a learned gate (§5.2), with tuned lens as diagnostic upper
 211 bound (§5.3).

212 5.1 Compound Error and Reasoning-Specific Amplification

213 **Under raw logit lens, question-level accuracy is 0% at every layer for all four models** (Table 53),
 214 regardless of exit strategy. Even at 89% depth with 73% token accuracy, at least one token per
 215 question is incorrect; evaluating only the final boxed answer, accuracy still drops from 84.3% to <2%
 216 at L32 (Appendix S).

217 Two levels of failure compound: *compound error* over $\sim 3,000$ tokens (even 99.2% per-token accuracy
 218 yields near-zero Q_{Acc} ; Table 50) and a reasoning-specific *per-token deficit* (-15.3 pp; Table 3). Even
 219 boxed-answer-only evaluation collapses to 1.6% (Appendix S).

220 CALM-style adaptive exit [Schuster et al., 2022] fares no better (Table 4): softmax criteria yield
 221 **0% Q_{Acc} across all thresholds**; cosine similarity at $\tau=0.97$ achieves 99.2% token accuracy yet 0%
 222 Q_{Acc} . Per-type optimized thresholds yield only +0.06% over uniform. Logistic probes achieve AUC
 223 0.78–0.87 (Table 54) but AUC *decreases* with depth—deployed as exit gates, they yield 23% Q_{Acc} at
 224 0.054% savings (Appendix X).

225 **Convergence hardens within traces.** Never-converging tokens increase as reasoning progresses;
 226 answer-segment tokens exhibit the highest non-monotonic rate despite converging earliest (Ap-
 227 pendix W, S).

228 5.2 Projection Bottleneck and Proof-of-Concept Recovery

229 A **+45.6pp gap** between raw (12.6%) and tuned (58.2%) oracle savings (Figure 4a) localizes the
 230 bottleneck to projection quality. As a proof-of-concept, we train a learned MLP gate on tuned-lens
 231 confidence features² to test whether this gap can be partially closed. The gate achieves **63.0%**
 232 Q_{Acc} **at 6.28% savings** (nested CV: outer fold trains tuned lens, inner fold trains gate; 9-layer exit
 233 grid—grid sensitivity in Table 6; Appendix P); net savings after probe overhead remain $\sim 5.3\%$
 234 (Appendix T.1). Margin and top- k mass drive exit decisions ($|\text{coef}| > 0.5$), while type identity
 235 contributes minimally ($|\text{coef}| < 0.02$; per-type gates: V2c 64%, 18.1% savings). Compound error
 236 remains the binding constraint: 63.0% vs. $\approx 91\%$ near-full-depth Q_{Acc} . Gate architecture choice is
 237 validated in Appendix P.1.

238 **Gap-closing progression.** Four tiers emerge: (i) raw CALM: 0% Q_{Acc} ; (ii) tuned-lens CALM
 239 ($\tau=0.99$): 29.4% Q_{Acc} , 11.4% savings; (iii) V2a gate ($\gamma=0.99$): 63.0% Q_{Acc} , 6.28% savings;
 240 (iv) oracle ceiling: 58.2% savings. Each tier closes a distinct component: projection quality (i $\rightarrow$ ii),
 241 learned boundaries (ii $\rightarrow$ iii), residual compound error (iii $\rightarrow$ iv).

242 **Controls and scale.** A matched non-reasoning control (Qwen2.5-7B-Instruct, $n=500$) confirms
 243 reasoning-specificity: gate val_loss plateaus at ≈ 0.50 (chance), yielding 5.7% savings at near-zero
 244 Q_{Acc} ; the V2a gate on R1-Qwen achieves 63.0% Q_{Acc} (val_loss well below chance), demonstrating
 245 that the learned exit signal is reasoning-specific even though absolute savings are modest. QwQ-32B
 246 ($n=500$) confirms CALM failure at 32B: $\tau=0.95$ exits zero tokens; $\tau=0.90$ collapses to 2.4% Q_{Acc} .
 247 The V2a gate, however, recovers $61.0\% \pm 2.4\%$ Q_{Acc} at $4.98\% \pm 0.44\%$ savings ($\gamma=0.99$, nested CV,
 248 MATH-500; Appendix V), demonstrating that learnable exit signals persist at 32B despite standard
 249 criteria failing; the lower savings reflect exit grid sparsity relative to the 64-layer depth. R1-Distill-
 250 Qwen-7B gate also fails (val_loss ≈ 0.50 , 4.5% savings; Appendix V). The Phi-4 CALM gap (2.8%
 251 vs. R1-Qwen’s 29.4%) is analyzed in §5.3.

252 **Cross-architecture V2a transfer.** To test whether the V2a gating signal generalizes beyond the
 253 training architecture, we evaluate the same gate on R1-Llama-8B ($n=500$, GSM8K), Phi-4-reasoning-
 254 14B ($n=500$, GSM8K), and QwQ-32B ($n=500$, MATH-500), all under nested CV. At $\gamma=0.99$:
 255 R1-Llama achieves $55.0\% \pm 8.9\%$ Q_{Acc} at $9.41\% \pm 0.62\%$ savings; Phi-4 achieves $48.0\% \pm 7.3\%$
 256 Q_{Acc} at $12.70\% \pm 0.71\%$ savings; QwQ-32B achieves $61.0\% \pm 2.4\%$ Q_{Acc} at $4.98\% \pm 0.44\%$ savings
 257 (lower savings reflect exit grid sparsity for 64 layers; Appendix V). Savings range from $\sim 5\%$ to 12.7%
 258 across architectures and scales—modest but consistently nonzero, versus CALM’s 0%. However, exit
 259 layer grid ablation (Table 6) shows that savings are grid-configuration-dependent: on the two models
 260 tested under both 9-layer and 7-layer grids, savings range from 4.1–6.3% (R1-Qwen) and 8.4–9.4%
 261 (R1-Llama). QwQ-32B ($\sim 5\%$) and Phi-4 (12.7%) are single-grid results not yet validated under
 262 alternative configurations. Validation loss separates the models: all three $\geq 8\text{B}$ reasoning models yield
 263 val_loss well below 0.50 (R1-Qwen $\ll 0.50$; Phi-4 ≈ 0.41), whereas R1-Distill-Qwen-7B (≈ 0.506 ;
 264 distilled from the same R1 teacher as R1-Llama) and the non-reasoning control (Qwen2.5-7B: ≈ 0.50)
 265 remain at chance level. Since R1-Llama (8B, distilled) succeeds while R1-Distill-Qwen (7B, same
 266 teacher) fails, capacity and/or architecture differences may contribute—though the current evidence
 267 (four successes vs. two failures) does not isolate the mechanism. Per-architecture calibration would
 268 recover efficiency (Appendix V).

269 **Nested CV and information leakage.** Non-nested evaluation (where a single tuned lens is trained
 270 on all data before gate cross-validation) inflates savings estimates: R1-Qwen savings drop from
 271 20.2% (non-nested) to 6.28% (nested), while Q_{Acc} decreases from 68.2% to 63.0%—the shared
 272 tuned-lens information between training and test sets created an optimistic exit signal. R1-Llama
 273 provides a stability check: savings are essentially unchanged (9.08% $\rightarrow$ 9.41%), consistent with

²For each cross-validation fold, affine transforms are fitted on the fold’s 400 training questions, ensuring no test-set information leaks into the tuned lens. Each transform has $d^2 + d$ parameters per layer (one affine transform per layer; $d=4096$ for 8B models).

Table 5: Effect decomposition on R1-Qwen3-8B (MATH-500, $\gamma=0.99$). Leakage (rows 1 $\rightarrow$ 2) accounts for 88% of the savings drop; exit-layer grid density (rows 2 $\rightarrow$ 3) accounts for 12%.

Condition	Q_{Acc}	Savings	Gate val_loss
Leaky, 7-layer	68.2%	20.2%	~ 0.35
Clean, 7-layer	75.8%	4.08%	~ 0.49
Clean, 9-layer	63.0%	6.28%	~ 0.46

Table 6: Exit layer grid sensitivity on R1-Qwen3-8B (MATH-500, 5-fold nested CV). A sparser grid yields higher Q_{Acc} at lower savings, but gate val_loss approaches chance level (≈ 0.49 vs. 0.46), indicating degraded gate learnability.

γ	Grid	Q_{Acc}	Savings	ΔQ_{Acc}
0.90	9-layer	$1.2\% \pm 0.8\%$	$21.5\% \pm 1.0\%$	—
	7-layer	$2.2\% \pm 1.3\%$	$18.1\% \pm 0.5\%$	+1.0 pp
0.95	9-layer	$8.2\% \pm 0.8\%$	$15.3\% \pm 0.9\%$	—
	7-layer	$12.6\% \pm 2.7\%$	$12.0\% \pm 0.3\%$	+4.4 pp
0.99	9-layer	$63.0\% \pm 5.2\%$	$6.3\% \pm 0.6\%$	—
	7-layer	$75.8\% \pm 2.8\%$	$4.1\% \pm 0.3\%$	+12.8 pp

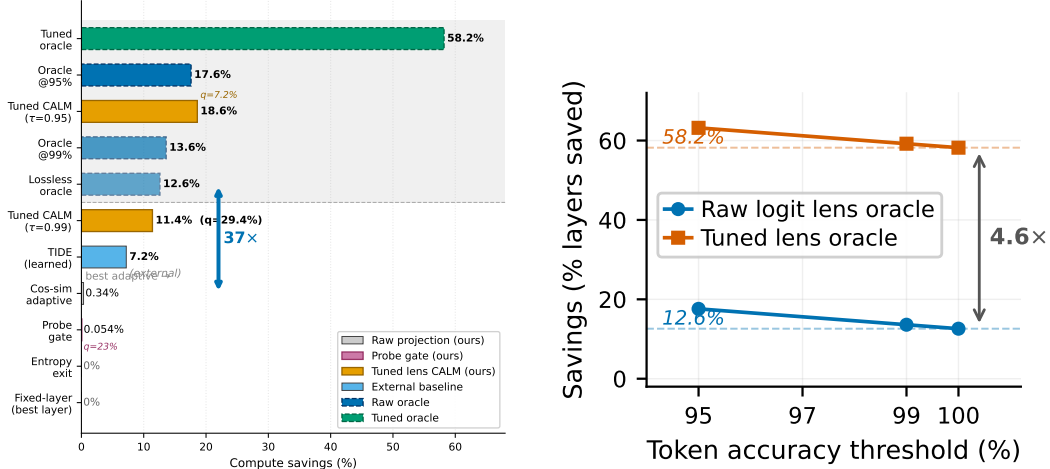
the cross-architecture transfer setup where the tuned lens was always trained on R1-Qwen data, eliminating same-architecture leakage. All numbers in this paper use nested CV unless otherwise noted.

Effect decomposition: leakage vs. exit-layer grid. A 7-layer ablation (exit layers {8, 12, 16, 20, 24, 28, 32}, nested CV) decomposes the savings decline into two separable effects (Table 5). Holding the grid constant at 7 layers, fixing leakage reduces savings by -16.1 pp (leaky 20.2% $\rightarrow$ clean 4.08%). Holding the evaluation clean and expanding from 7 to 9 exit layers recovers only $+2.2$ pp (4.08% $\rightarrow$ 6.28%). The two effects sum to -13.9 pp, matching the total observed decline. Leakage thus accounts for 88% of the savings drop (16.1 of 18.3 pp), while exit-layer grid density explains only 12% (2.2 pp). Gate val_loss confirms the pattern: it degrades from ~ 0.35 (leaky) to 0.46–0.49 (clean), approaching chance level (0.50)—the leaky evaluation allowed the gate to exploit shared tuned-lens information rather than learn a genuine exit criterion.

Exit layer grid sensitivity. The accuracy–savings tradeoff depends on the exit layer grid (Table 6). Reducing from 9 to 7 exit layers consistently increases Q_{Acc} while decreasing savings across all γ thresholds. However, gate val_loss reveals that this shift reflects degraded gate learnability rather than intentional conservatism: the 9-layer gate achieves val_loss ≈ 0.46 (well below chance at 0.50), while the 7-layer gate reaches only ≈ 0.49 (near chance). The sparser grid restricts the gate’s learning signal, causing it to default toward not exiting—the higher Q_{Acc} is a byproduct of gate failure rather than a learned conservative strategy. Per-fold variance decreases under the 7-layer grid (std 2.8% vs. 5.2% at $\gamma=0.99$), consistent with less discriminative gating.

The Q_{Acc} gap widens at higher γ : from $+1.0$ pp at $\gamma=0.90$ to $+12.8$ pp at $\gamma=0.99$. At aggressive thresholds ($\gamma \leq 0.95$), most tokens exit early under both grids and the additional exit points matter little. At conservative thresholds ($\gamma=0.99$), the gate is selective enough that the reduced exit layer set changes the effective decision boundary. R1-Llama shows a directionally consistent but smaller effect ($+7.4$ pp ΔQ_{Acc} , -1.05 pp savings; 95% CIs overlap, so the cross-model trend is directionally consistent rather than statistically confirmed; Appendix P.2). The exit layer grid is thus a sensitive design choice that substantially affects both accuracy and gate learnability—an effect magnified in reasoning models, where compound error amplifies small changes in per-token exit decisions across thousand-token traces. Practitioners must validate gate quality (via val_loss) when adjusting the exit layer configuration. The token-level exit patterns and representation maturity findings (§4–§4.4) are computed at individual layers and are independent of which layers are designated as exit points; the grid sensitivity documented here applies specifically to the V2a gating results.

While concurrent system-level approaches such as TIDE achieve throughput gains by learning per-token exit routers, our results suggest that without addressing the underlying representation



(a) Raw vs. tuned oracle gap (+45.6pp). (b) Oracle savings vs. token acc. Tuned lens: Figure 4: End-to-end exit analysis on R1-Qwen3-8B (300k) → 58.2%. (a) The oracle gap (+45.6pp). (b) Tuned lens raises the oracle ceiling from 12.6% to 58.2%. This diagnostic analysis quantifies the projection bottleneck and its impact on early-exit feasibility.

immaturity (the +45.6 pp tuned lens gap), such routers risk silent accuracy degradation on reasoning tasks. TIDE’s 98–99% task accuracy reflects a more tolerant metric (final-answer correctness) that absorbs intermediate token errors; our Q_{Acc} —requiring *all* tokens to match full-depth output—reveals the per-token deficit that task accuracy masks.

5.3 Tuned Lens as Diagnostic Upper Bound

Tuned lens achieves non-zero question accuracy: a token exits at the shallowest layer where tuned-lens confidence exceeds τ (per-layer affine transforms from §4.2; nested CV, 400 train / 100 test per fold; Table 4).

Question accuracy reaches **29.4% at $\tau = 0.99$** (7.2% at $\tau \geq 0.95$), demonstrating that improved projection alone partially overcomes the per-token deficit. The oracle curve (Figure 4b) confirms projection quality as the bottleneck: tuned lens raises lossless savings from 12.6% to 58.2% (+45.6pp; generalizes to HumanEval, Appendix Q).

Cross-difficulty transfer. On MATH-500, tuned lens oracle drops to 35.4% (−22.8pp) while raw oracle remains stable, confirming the gap resides in the learned projection. **In-distribution retraining recovers it:** MATH-500 tuned oracle reaches 57.1% (+46.7pp), nearly matching GSM8K (58.2%)—distribution mismatch, not intrinsic difficulty. Despite oracle recovery, CALM Q_{Acc} remains at 18% (~13% savings), confirming compound error as the binding constraint (Appendix R; GSM8K replication in Appendix Z).

Cross-architecture and scale validation. R1-Llama confirms at $\tau=0.99$: 52.0% Q_{Acc} (9.3% savings; Table 4); oracle gap +36.9pp. Phi-4 (14B) shows +49.3pp oracle gap (27.3%→76.6%; Table 43), though tuned-lens CALM achieves only 2.8% Q_{Acc} (Appendix T). Phi-4’s tuned-lens CALM Q_{Acc} of 2.8% at $\tau=0.90$ (vs. R1-Qwen’s 29.4% at $\tau=0.99$) likely reflects two compounding factors: (1) Phi-4’s traces are $\sim 5\times$ longer ($\sim 16\text{K}$ vs. $\sim 3\text{K}$ tokens), amplifying compound error exponentially, and (2) a pronounced penultimate-layer representation shift (86.4% of tokens show a cosine dip at L38→L39; Appendix V) degrades CALM’s confidence signal at the layer where most tokens would exit. Despite the larger oracle gap (+49.3 pp), CALM’s threshold-based criterion cannot leverage this recoverable information. The projection bottleneck generalizes across all three families (+36.9–49.3pp); QwQ-32B and R1-Distill-Qwen-7B in §5.2. The V2a gate confirms cross-architecture transfer (48–63% Q_{Acc} at ~ 5 –10% savings (grid-dependent) under nested CV, $\gamma=0.99$; §5.2).

6 Conclusion

Two separable failure patterns—oscillatory dynamics (distillation-specific) and representation immaturity (reasoning-general)—are confirmed via thinking-mode ablation and at 14B scale; CALM failure persists at 32B. Tuned lens narrows the Reasoning Early Exit Gap by +36.9–49.3pp across three architecture families; a learned gate achieves modest recovery (48–63% Q_{Acc} at ~5–10% savings (grid-dependent) under nested CV; R1-Qwen, R1-Llama, Phi-4, QwQ-32B; 8B–32B), highlighting the fundamental depth of the Reasoning Early Exit Gap; the 7B distilled variant remains at chance-level val_loss and compound error is the binding constraint.

Implications. The taxonomy’s value is diagnostic rather than predictive—type identity contributes minimally to exit decisions ($|\text{coef}| < 0.02$; Appendix P). Segment-level methods [Yang et al., 2025a] and hierarchical verification are promising paths to close the oracle-to-deployment gap.

7 Limitations

Both distilled models share one teacher (DeepSeek-R1); cross-architecture V2a transfer tested on three non-Qwen-8B models (R1-Llama 8B, Phi-4 14B, QwQ-32B). Scale: 8B–32B full V2a analysis with **reversal at 1.5B** (Appendix V); QwQ-32B includes V2a gate evaluation but no tuned lens training or oracle analysis, so the projection bottleneck’s magnitude at 32B is unknown. QwQ-32B uses bfloat16 accumulation (vs. float32 at 8B–14B) and answer_length_tercile stratification (vs. difficulty_level); see Appendix V. Domain: math/code only. The gate’s capacity/architecture threshold (four successes vs. two failures) rests on limited evidence; additional architectures and scales are needed to isolate the mechanism.

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478 A Taxonomy Robustness Validation

479 We assess the robustness of the four-category token role taxonomy (§3.2) along three axes: keyword
480 set sensitivity, boundary ambiguity, and unsupervised clustering structure.

481 **Reasoning keyword list.** The 28 reasoning keywords are matched via case-insensitive regex on
482 decoded token strings: *therefore, because, so, implies, let, assume, suppose, hence, thus, since, given,*
483 *if, then, means, conclude, verify, check, consider, note, recall, know, think, wait, actually, but, however,*
484 *instead, alternatively.* Tokens matching any keyword are classified as reasoning keywords unless
485 they also match the numerical or special category (which take priority).

486 **Keyword perturbation.** To test sensitivity to this keyword set, we randomly replace 5 of 28
487 keywords per trial (10 seeds). Table 7 shows that numerical, special, and other rates are invariant
488 (by construction), while the reasoning keyword rate varies from 0.2% to 5.3% (mean 3.6%, baseline
489 4.2%). The ~ 5 pp range reflects boundary words (e.g., “but”, “let”) that straddle reasoning and
490 general usage. Crucially, the *ranking* of non-monotonicity rates across types is preserved under all
491 perturbations: numerical $\gg$ special $>$ other $>$ reasoning keyword.

492 **Boundary ambiguity.** We analyze tokens at category boundaries—those whose classification could
493 change under reasonable alternative rules. Reclassifying all ambiguous reasoning keywords as “other”
494 shifts the reasoning keyword rate from 4.19% to 0.16% (-4.03 pp) while changing the other rate by
495 only -0.41 pp. Reclassifying special-boundary tokens as numerical shifts both rates by <1.5 pp. In
496 all worst-case scenarios, the cross-type non-monotonicity ranking is preserved.

Table 7: Non-monotonicity rate (%) under keyword set perturbation (10 seeds, 5 swaps each). Numerical and special rates are unaffected; reasoning keyword rate ranges from 0.2% to 5.3%.

	Numerical	Special	Reasoning KW	Other
Baseline	42.93	27.35	4.19	11.75
Mean (10 seeds)	42.93	27.35	3.64	11.70
Min	42.93	27.35	0.21	11.52
Max	42.93	27.35	5.32	11.77

Table 8: Enrichment ratios (observed/expected) for k -means clusters ($k=3$) over convergence trajectories. Values >1 indicate over-representation. Non-mono rate is within-cluster non-monotonicity.

Cluster (non-mono rate)	Numerical	Special	Reasoning KW	Other
Low-conv (24.2%)	0.98	1.14	1.20	0.93
Mid-conv (21.1%)	1.03	0.81	0.50	1.12
Early-conv (9.7%)	0.97	1.25	2.19	0.80

Unsupervised clustering structure. We apply k -means ($k=3$) to per-token convergence trajectories (cosine similarity curves across layers) without using the regex taxonomy labels. Table 8 reports the enrichment ratio (observed/expected) of each regex category in each cluster. The low-convergence cluster (highest non-monotonicity, 24.2%) is enriched for special tokens ($1.14\times$) and reasoning keywords ($1.20\times$), while the mid-convergence cluster (21.1%) is enriched for other tokens ($1.12\times$) and numerical tokens ($1.03\times$). Reasoning keywords are strongly enriched ($2.19\times$) in the early-convergence cluster (lowest non-monotonicity, 9.7%), consistent with their smooth but immature convergence pattern (representation immaturity) identified in §4.3. Note that overall NMI (Normalized Mutual Information; 0.014) and ARI (Adjusted Rand Index; 0.032) between clusters and token roles are low, indicating that token role and convergence dynamics are largely orthogonal dimensions—the enrichment patterns above reflect relative tendencies, not categorical alignment. Formal predictive validation is provided by logistic regression (Appendix Y).

Question subset stability. To rule out single-sample artifacts, we run the R1-Qwen H1 analysis on three disjoint 100-question GSM8K subsets (questions 0–99, 100–199, 200–299; greedy decoding, no randomness beyond question selection). Table 9 reports non-monotonicity rates at three thresholds.

The per-type non-monotonicity ranking is identical across all three subsets: numerical $\gg$ special $>$ other $>$ reasoning keyword (at $\theta=0.03$, numerical ranges 32–36%, reasoning keyword 2.8–3.1%). Layer-wise dip concentration is also stable: layers 16–20 account for 66–71% of all cosine similarity dips across subsets.

H2: Exit accuracy stability. Table 10 reports logit lens exit accuracy (token-level agreement with the final layer) across the same three subsets ($n=50$ per subset, sampled from each 100-question pool).

Rate reconciliation. The subset-level H1 rates ($\sim 12\%$ at $\theta=0.03$) are lower than the full-sample rate (20.8%) reported in §4.1. This difference reflects dataset composition: the full sample pools GSM8K and MATH-500, and MATH-500 questions—which involve harder multi-step reasoning and longer traces—exhibit higher non-monotonicity. The stability check uses GSM8K subsets only to isolate question-sampling variance from dataset-difficulty effects.

Non-monotonicity threshold sensitivity. The default threshold $\theta=0.03$ is chosen to exceed noise-level fluctuations while remaining sensitive to meaningful dips. To verify that the reasoning $>$ base non-monotonicity gap is not an artifact of this choice, we sweep $\theta \in \{0.01, 0.02, 0.03, 0.05, 0.10\}$ across all three model pairs. Table 11 reports the pairwise delta (reasoning NM% – base NM%, in percentage points) at each threshold.

The delta peaks around $\theta \in [0.02, 0.03]$ and decays at both extremes—at very low θ , noise-level fluctuations inflate both rates; at very high θ , few tokens in either model exceed the threshold.

Table 9: Non-monotonicity rate (%) across three disjoint GSM8K subsets ($n=100$ each, R1-Qwen3-8B). Max cross-subset $\Delta < 1\text{pp}$ at all thresholds.

θ	Subset 0–99	Subset 100–199	Subset 200–299	Max Δ
0.01	31.3	31.9	31.8	0.6pp
0.03	11.7	12.2	12.2	0.5pp
0.05	4.3	4.5	4.4	0.2pp

Table 10: Logit lens exit accuracy (%) across three disjoint GSM8K subsets ($n=50$ per subset, R1-Qwen3-8B). Max $\Delta < 1.1\text{pp}$ at all layers.

Layer	Subset 0–99	Subset 100–199	Subset 200–299	Max Δ
L28	23.9	24.4	24.4	0.5pp
L32	49.7	50.8	50.5	1.1pp
L34	74.9	75.4	75.6	0.7pp

Crucially, no sign reversal occurs at any threshold for the 8B or 14B pairs, confirming that the core claim is robust to threshold selection.

B Bootstrap Confidence Intervals

All confidence intervals are computed via question-level bootstrap resampling (1,000 iterations): we resample questions with replacement, recompute the metric over all tokens in the resampled set, and report the 2.5th/97.5th percentiles. This approach accounts for within-question token correlation that would be missed by token-level resampling.

All intervals are narrow: the widest is the Llama exit accuracy CI ($\pm 1.6\text{pp}$), reflecting the smaller sample size (300 vs. 1,114 questions). The non-overlapping CIs for R1 numerical (41.6–44.2%) vs. Llama numerical (0.9–2.6%) confirm that the cross-model non-monotonicity reversal (§4.1) is statistically robust. The sharp jump at $\tau \in [0.6, 0.7]$ is tightly estimated at $+44.82\text{pp} \pm 0.5\text{pp}$.

C 2×2 ANOVA Details

We report the full balanced two-way ANOVA (4×100 questions, $\theta=0.03$). As noted in §3.1, this is an observational study; η^2 values quantify strength of association, not causal effect sizes. Each cell contains 100 questions; the dependent variable is per-question non-monotonicity rate. Table 13 shows main effects, interaction, and group means.

The interaction ($\eta^2=0.096$) reflects asymmetric sensitivity to reasoning training: Llama’s non-monotonicity rate more than triples under reasoning training ($4.4\% \rightarrow 14.3\%$, $3.27\times$), while Qwen’s rises modestly ($10.6\% \rightarrow 12.4\%$, $1.17\times$). This may reflect differences in non-reasoning model convergence dynamics: Llama’s low baseline (4.4%) leaves more room for reasoning training to disrupt monotonic convergence.

Comparison with unbalanced analysis. Table 15 compares the balanced (4×100) and unbalanced (full-sample) ANOVA results. All effects remain significant in both analyses, but balanced η^2 values are uniformly larger (e.g., training: 0.285 vs. 0.157). This confirms that sample-size imbalance *dilutes* effect sizes in the unbalanced analysis (R1-Qwen’s 1,114 questions dominate); the balanced analysis thus provides a more accurate estimate.

Assumption checks. Shapiro-Wilk rejects normality and Levene’s test rejects homogeneity of variance, as expected for bounded proportions. Welch ANOVA ($F=222.35$, $p < 10^{-50}$), Kruskal-Wallis ($H=221.55$, $p < 10^{-48}$, $\eta_H^2=0.552$), and a permutation test ($p=0.0001$, 10^4 permutations) all confirm the reported effects are robust to these violations.

η^2 vs. η_H^2 discrepancy. The Kruskal-Wallis $\eta_H^2=0.552$ is approximately $2\times$ the parametric $\eta^2=0.285$. This gap arises because η_H^2 is computed on ranks, amplifying signal from bimodal

Table 11: Pairwise non-monotonicity delta (reasoning – base, in pp) across thresholds. All deltas are positive: the reasoning > base pattern holds at every θ .

Model Pair	$\theta=0.01$	$\theta=0.02$	$\theta=0.03$	$\theta=0.05$	$\theta=0.10$
8B-Qwen (R1 vs. base)	+8.9	+10.8	+9.2	+5.4	+1.0
8B cross-arch [†] (R1-Qwen vs. Llama)	+2.8	+10.9	+12.3	+9.1	+1.7
14B-Phi (reasoning vs. base)	+17.5	+14.8	+14.1	+9.7	+1.0

[†]Compares R1-Qwen3-8B (distilled reasoning) vs. Llama-3.1-8B-Instruct (non-reasoning); delta reflects both training paradigm and architecture differences.

Table 12: Bootstrap 95% confidence intervals for core metrics (1,000 question-level resamples).

Metric	Point	CI Lower	CI Upper	n
H1: Numerical non-mono rate	42.93%	41.58%	44.20%	1,114 q
H1: Reasoning KW non-mono rate	4.19%	4.03%	4.35%	1,114 q
H1: Cross-model reversal (R1 num)	42.93%	41.58%	44.20%	1,114 q
H1: Cross-model reversal (Llama num)	1.60%	0.93%	2.62%	300 q
Phase transition jump (τ : 0.6→0.7)	+44.82pp	+44.33pp	+45.31pp	500 q
H2: R1 exit acc (L28)	32.41%	32.09%	32.71%	498 q
H2: Llama exit acc (L28)	74.30%	72.78%	76.01%	300 q

distributions (reasoning keyword tokens cluster near 0% and 100% non-monotonicity); the rank transform effectively linearizes this bimodality, yielding a larger explained-variance fraction. Both estimates agree that training condition dominates architecture, so the directional conclusion is robust; only the quantitative effect size depends on the choice of estimator.

Subsampling stability. To verify that η^2 values are not driven by a particular random subset, we repeat the balanced ANOVA (4×100) with 10 independent random draws (different 100-question subsets from cells with >100 questions). Results: training $\eta^2 = 0.277 \pm 0.006$, architecture $\eta^2 = 0.108 \pm 0.002$, interaction $\eta^2 = 0.107 \pm 0.015$ (mean $\pm$ sd). All standard deviations are below 0.02, confirming the reported effect sizes are stable across subsamples.

Alternative statistical models. To verify findings under distributional assumptions more appropriate for bounded proportions, we fit (i) a beta regression and (ii) a quasibinomial GLM on the same 4×100 question-level non-monotonicity rates. A Kolmogorov–Smirnov test on empirical NM rates confirms consistency with a Beta distribution ($p=0.56$). Both models agree with ANOVA in direction and significance: beta regression yields a training coefficient of 0.668 ($p=2.2 \times 10^{-30}$) and quasibinomial GLM yields 0.590 ($p=7.4 \times 10^{-31}$), both indicating reasoning > non-reasoning. Architecture effects are also significant in both models ($p<0.001$). The three approaches—ANOVA, beta regression, and quasibinomial GLM—reach identical directional conclusions, confirming that the reported associations are not artifacts of the ANOVA normality assumption. Note that the main-text ANOVA operates on question-level aggregates ($N=400$), while the logistic regression in Appendix Y operates on individual tokens ($N \approx 4.6\text{M}$); differences in effect magnitude between the two granularities are expected and do not affect directional conclusions.

D Cross-Correlation Analysis

We examine the relationship between non-monotonic convergence (H1) and exit accuracy (H2) at the question level. For each of the 500 R1-Qwen questions, we compute: (1) the fraction of tokens with non-monotonic trajectories, and (2) the mean cosine similarity to the final layer at each checkpoint (a proxy for exit accuracy).

At L16, questions with more non-monotonic tokens have *higher* convergence ($\rho = +0.44$, $p < 10^{-24}$)—these are questions where representations are actively reorganizing, producing temporarily high similarity before diverging. By L24–L28, the correlation flips to weakly negative ($\rho = -0.09$ to -0.14), indicating that non-monotonic questions achieve slightly *lower* convergence at deeper layers. However, the magnitude is small: the top-25% most non-monotonic questions have only 0.44pp lower L28 convergence than the bottom-25%.

Table 13: Balanced 2×2 ANOVA on question-level non-monotonicity rate ($N=400$, 100 per cell).

Effect	$F_{1,396}$	p	SS	η^2	Interpretation
Training (reasoning vs. non-reasoning)	283.78	2.1×10^{-48}	0.494	0.285	Strongest association
Architecture (Qwen vs. Llama)	110.65	5.5×10^{-23}	0.193	0.111	Strong association
Training $\times$ Architecture	95.34	2.5×10^{-20}	0.166	0.096	Significant interaction

Table 14: Group means (question-level non-monotonicity rate) and pairwise contrasts.

	Qwen	Llama	Δ (Llama – Qwen)
Reasoning	12.4%	14.3%	+1.9pp
Non-reasoning	10.6%	4.4%	−6.2pp
Δ (reasoning – non-reasoning)	+1.8pp	+9.9pp	—
Ratio (reasoning / non-reasoning)	1.17 $\times$	3.27 $\times$	—

Per-type breakdown. Table 17 reveals that the cross-correlation structure varies sharply by token role. Numerical tokens show a consistent negative correlation ($\rho = -0.244$, $p < 10^{-7}$): questions with more non-monotonic numerical tokens achieve lower convergence, consistent with oscillation disrupting final-layer stability. Reasoning keywords show the opposite pattern ($\rho = +0.427$, $p < 10^{-23}$): questions with more non-monotonic reasoning keywords actually achieve *higher* convergence—a counterintuitive result suggesting that some reasoning keyword non-monotonicity reflects productive representation refinement rather than harmful oscillation.

E Chain Length Analysis

Reasoning traces vary widely in length (274–16,384 tokens, median 2,877). We test whether chain length confounds the token-type effect on non-monotonicity.

Per-dataset correlations (critical caveat). The aggregate correlation between chain length and non-monotonicity ($\rho = 0.151$, $p < 10^{-8}$, $n = 1,466$) conceals opposite within-dataset relationships (Table 18). GSM8K shows a *negative* correlation ($\rho = -0.207$): longer GSM8K traces have *lower* non-monotonicity, possibly because harder questions require more straightforward step-by-step computation. MATH-500 shows a weak positive correlation ($\rho = 0.092$). The positive aggregate ρ is an **aggregation artifact**: MATH-500 questions are both longer (median 5,746 vs. 2,166 tokens) and more non-monotonic (mean 21.4% vs. 11.9%), creating a spurious between-dataset association.

Quintile analysis. Table 19 shows that non-monotonicity rates are relatively stable across chain-length quintiles within the pooled data, with only the longest quintile ($> 6,645$ tokens) showing a notable increase—associated with the higher proportion of MATH-500 questions in that bin.

Token role dominates chain length. Within the same question, numerical tokens average 27.8pp higher non-monotonicity than reasoning keywords (paired $t = 90.0$, $p \approx 0$, $n = 1,466$). Chain length modulates this gap ($\rho = 0.341$ between chain length and the numerical–reasoning keyword difference, $p < 10^{-40}$), indicating a partial confound. However, the gap is positive in *every* quintile across both datasets, confirming that token role is the primary correlate.

Per-type correlations with chain length are also type-specific: numerical tokens show a weak positive trend ($\rho = 0.192$, $p < 10^{-13}$), while reasoning keywords show a strong *negative* trend ($\rho = -0.650$, $p < 10^{-176}$)—longer chains have proportionally *fewer* non-monotonic reasoning keywords, likely because longer traces contain more routine reasoning connectives that converge easily.

Summary. The aggregate chain-length correlation is confounded by dataset composition and should not be interpreted causally—a form of Simpson’s paradox also discussed in Appendix J. Within each dataset, the correlation is weak ($|\rho| \leq 0.21$), and the token-role effect (27.8pp numerical–reasoning keyword gap) dwarfs the chain-length effect at every quintile. All main-text claims about token role are robust to chain-length controls.

Table 15: Balanced vs. unbalanced ANOVA comparison. Balanced η^2 is uniformly larger, indicating the unbalanced analysis underestimates effect sizes.

Effect	Balanced (4×100)		Unbalanced (full)	
	F	η^2	F	η^2
Training	283.78	0.285	192.52	0.157
Architecture	110.65	0.111	75.07	0.061
Interaction	95.34	0.096	90.78	0.074

Table 16: Spearman ρ between per-question non-monotonicity rate and convergence at each layer (R1-Qwen, $n=500$). The sign flip from positive (L16) to negative (L24–L28) indicates that non-monotonic questions *catch up* by deeper layers.

	L16	L20	L24	L28
Spearman ρ	+0.437	+0.067	−0.141	−0.091
p -value	8.9×10^{-25}	0.135	0.002	0.042

629 F Per-Type Exit Accuracy Tables

630 We report three complementary views: (1) overall exit accuracy at multiple layers for all four models,
631 (2) per-type breakdown at the deepest sub-final layer, and (3) proxy validation across models.

632 **Overall exit accuracy depth profile.** Table 20 shows overall token-level exit accuracy at each
633 checkpoint layer. Sample sizes: R1-Qwen (500q), R1-Llama (100q), Qwen3-8B non-reasoning
634 (100q), Llama (200q). A sharp divergence is evident: all models remain below 5% through L16, then
635 separate sharply.

636 **Per-type breakdown.** Table 21 provides per-type exit accuracy at the deepest sub-final layer for all
637 four models. Sample sizes: R1-Qwen (500q), R1-Llama (100q), Qwen3-8B non-reasoning (100q),
638 Llama (200q, GSM8K [100:300]).

639 Several cross-model patterns emerge: (1) Reasoning training compresses the per-type range at L28:
640 both reasoning models show narrow spreads (R1-Qwen: 5.2pp, R1-Llama: 5.9pp) vs. wide spreads
641 in non-reasoning models (Qwen3-8B (non-reasoning): 26.3pp, Llama: 26.1pp). This compression
642 is consistent across architectures. (2) In non-reasoning models, reasoning keywords achieve the
643 *highest* exit accuracy (Qwen3-8B (non-reasoning): 60.5%, Llama: 79.4%), but reasoning training
644 is associated with the largest reduction (−28.8pp for Qwen). In R1-Llama, reasoning keywords
645 are the *lowest* type (65.8%), confirming the cross-architecture pattern. (3) The L28→L32 ranking
646 reversal in R1-Qwen (numerical rises from worst to best) confirms the dual failure pattern: numerical
647 tokens oscillate at intermediate layers but ultimately converge, while reasoning keywords converge
648 monotonically but insufficiently.

649 **Proxy validation.** Table 22 reports the cosine-similarity proxy Spearman ρ across models.

650 R1-Llama shows moderate overall ρ at L24 (0.319), but this aggregate masks a sharp per-type
651 divergence: reasoning keyword tokens exhibit $\rho = -0.515$ at L24—the proxy is *inversely* misleading
652 for these tokens, with higher cosine similarity corresponding to *lower* exit accuracy. This per-type
653 inversion (Table 23) is worse than uninformative: a practitioner relying on cosine similarity would
654 preferentially exit the wrong reasoning keyword tokens while correctly routing numerical tokens
655 ($\rho=0.307$).

656 G Oracle Savings Details

657 The oracle upper bound (§5.2) assigns each token to its shallowest correct exit layer from the candidate
658 set $\{4, 8, 12, 16, 20, 24, 28, 32, 34\}$. Exit accuracy at each layer is measured via LM head projection
659 (L8–L28 from the exit accuracy measurement; L32 from the full-depth evaluation; L34 interpolated
660 from L32=0.616 and L36=1.0). We report two variants: the *cosine oracle* (tokens can exit at any

Table 17: Per-type Spearman ρ between non-monotonicity rate and L28 convergence ($n=500$ questions).

	Numerical	Special	Reasoning KW	Other
Spearman ρ	-0.244	-0.185	+0.427	+0.079
p -value	3.4×10^{-8}	3.2×10^{-5}	1.3×10^{-23}	0.076

Table 18: Chain length vs. non-monotonicity rate, split by dataset. The aggregate positive correlation is an artifact of between-dataset confounding; within each dataset the relationship is weak or negative.

Dataset	n	Spearman ρ	p	Med. len	Mean non-mono	SD
GSM8K	1,000	-0.207	4.2×10^{-11}	2,166	11.9%	5.8%
MATH-500	466	+0.092	0.048	5,746	21.4%	8.4%
Aggregate	1,466	+0.151	6.3×10^{-9}	2,877	15.0%	8.0%

layer where they are correct) and the *nested bound* (tokens must be correct at all layers $\geq$ their exit layer, preventing oscillation-induced errors).

The cosine oracle and nested bound differ by only 0.5pp (19.8% vs. 19.2% at 95%), indicating that oscillation-induced false exits are a minor issue at the oracle level—the primary bottleneck is the fraction of tokens with mature representations. Reasoning keywords achieve the lowest savings (17.9%) due to their slow convergence, while “other” tokens (natural language filler) save the most (20.5%), consistent with their less demanding computational role.

Oracle grid resolution and discretization bias. Our oracle analysis uses a 9-layer grid (layers 4, 8, 12, 16, 20, 24, 28, 32, 36 out of 36 total), covering 25% of possible exit points. This introduces a systematic *discretization bias*: tokens that could exit at an intermediate layer (e.g., L10, L14) between grid points are forced to wait until the next checkpoint, underestimating true oracle savings. The bias is strictly downward—the 9-layer grid can never overestimate savings relative to a finer grid—making our reported oracle savings (12.6% raw, 58.2% tuned) conservative lower bounds. A finer grid would increase oracle savings but would also amplify compound error in deployment, as each additional exit point introduces another potential failure mode. We chose 9 layers to balance computational cost of hidden-state extraction (~ 8 GPU-hours for 500 questions $\times$ 36 layers) against resolution; the 4-layer spacing ensures coverage of early (L4–L12), middle (L16–L24), and late (L28–L36) representation stages.

Comparison with v1. An earlier analysis (v1) overestimated oracle savings at 21.1% due to using a logistic-fitted L32 accuracy of 78.0% rather than the measured 61.6%. The corrected v2 analysis uses measured values throughout; the 1.4pp reduction (21.1% $\rightarrow$ 19.8%) reflects the lower fraction of tokens that can exit before the final layers.

H Cross-Model Probe Comparison

Table 25 compares logistic regression probe AUC between R1-Qwen3-8B (reasoning-distilled, 500 questions) and Qwen3-8B non-reasoning (100 questions). Both use identical probing pipelines: PCA reduction to 128 dimensions, question-level train/test split (80/20).

Numerical tokens are the only type where the non-reasoning model does *not* have higher AUC (-0.013), likely because numerical oscillation (§4.3) creates distinctive hidden-state signatures that are easy to detect regardless of training condition. Reasoning keywords show the largest gap ($+0.169$): in the non-reasoning model, reasoning scaffolding tokens converge smoothly and predictably, making exit correctness easy to discriminate. Reasoning training is associated with convergence insufficiency (§4.2) that co-occurs with reduced discriminative signal precisely for these tokens.

The sample-size asymmetry (500 vs. 100 questions) means the non-reasoning model confidence intervals are wider; the consistent direction across all layers and three of four types suggests the gap is robust.

Table 19: Non-monotonicity rate by chain-length quintile ($n=1,466$ questions, pooled across datasets).

Quintile	Token range	Mean non-mono rate	SD
Q1 (shortest)	274–1,190	13.4%	5.4%
Q2	1,192–2,194	14.6%	7.2%
Q3	2,196–3,744	13.7%	7.4%
Q4	3,747–6,640	14.3%	8.5%
Q5 (longest)	6,645–16,384	18.8%	9.6%

Table 20: Overall token-level exit accuracy (%) at each checkpoint layer across the 2×2 design. L28 = 78% depth for Qwen (36 layers), 88% for Llama (32 layers).

Model (n questions)	L8	L12	L16	L20	L24	L28
R1-Qwen3-8B (500)	0.5	1.3	1.5	3.7	17.4	32.4
R1-Llama (100)	0.1	0.2	3.5	17.0	38.4	67.9
Qwen3-8B (100)	0.7	1.7	1.5	3.5	29.8	47.7
Llama-3.1-8B (200)	0.3	0.3	4.7	25.9	55.9	74.3

I OOD Truncation Analysis

48 of 164 HumanEval questions (29.3%) reached the 8,192-token generation limit. We verify that truncation does not distort the OOD findings by comparing the full, truncated, and non-truncated subsets.

Truncated questions show consistently lower non-monotonicity and slightly lower exit accuracy, but all differences are small (<5 pp for H1, <2 pp for H2) and not statistically significant. The per-type exit accuracy ranking at L28 is preserved in both subsets. Token-role distributions are nearly identical (numerical: 9.1% vs. 9.3%; reasoning keyword: 5.3% vs. 6.3%). We conclude that truncation does not materially affect the OOD generalization findings reported in Appendix Q.

J Simpson’s Paradox Analysis

Token-level and question-level non-monotonicity rates diverge across models because token counts vary by up to $13\times$ across models (Llama: ~ 295 tokens/question vs. Qwen3-8B (non-reasoning): $\sim 3,847$), reflecting both architecture and generation behavior. A single Qwen3-8B non-reasoning question contributes as many tokens as ~ 13 Llama questions, so token-level aggregation is dominated by the longer-generating models—a form of Simpson’s paradox.

The token-level rates suggest R1-Qwen (20.8%) has $\sim 2.4\times$ the non-monotonicity of Llama (8.5%). But at the question level, the gap narrows: 12.4% vs. 4.4% ($2.8\times$). Critically, the *ranking* is preserved under both aggregations: reasoning models show higher non-monotonicity regardless of weighting.

A closely related aggregation artifact appears in the chain-length analysis (Appendix E): the aggregate chain length–non-monotonicity correlation is *positive* ($\rho = +0.151$), but within each dataset the relationship is negative (GSM8K: $\rho = -0.207$) or negligible (MATH-500: $\rho = +0.092$). The positive aggregate arises because MATH-500 questions are both longer and more non-monotonic than GSM8K questions, creating a spurious between-dataset association—a textbook instance of Simpson’s paradox.

The balanced ANOVA (Appendix C) uses question-level rates with equal sample sizes (4×100), eliminating both sources of bias. All statistical claims in the main text are based on the balanced question-level analysis.

K Broader Impact

This work provides a diagnostic characterization of internal convergence dynamics in reasoning LLMs; it does not introduce new models, training methods, or deployed systems. The primary

Table 21: Per-type token-level exit accuracy (%) at the deepest sub-final layer and, where available, at a deeper checkpoint. Range = max – min across types. L28 = 78% depth for Qwen (36L), 88% for Llama (32L).

Model	Layer	Numerical	Special	Reas. KW	Other	Range
R1-Qwen3-8B	L28	28.5	31.7	31.7	33.7	5.2pp
	L32	65.3	60.0	55.2	61.7	10.1pp
R1-Llama	L28	71.7	66.2	65.8	68.4	5.9pp
Qwen3-8B	L28	34.2	49.4	60.5	49.8	26.3pp
	L32	67.4	74.2	77.8	74.2	10.4pp
Llama-3.1-8B	L28	76.7	53.3	79.4	79.0	26.1pp

Table 22: Cosine-similarity proxy validation: overall Spearman ρ between per-token cosine similarity (to final layer) and binary exit correctness at each layer.

Model	L12	L16	L20	L24	L28
R1-Qwen3-8B	0.005	0.015	0.032	0.034	0.050
R1-Llama	0.052	0.156	0.217	0.319	0.126
Qwen3-8B	—	—	—	0.20	0.15
Llama-3.1-8B	0.052	0.162	0.156	0.142	0.056

societal relevance is informing future efficiency methods: understanding why early exit fails on reasoning traces can guide the development of more compute-efficient inference, reducing the energy cost of deploying reasoning-capable models.

Early exit methods applied to safety-critical reasoning risk silently degrading answer quality; practitioners should validate end-to-end task accuracy, not just token-level metrics. In safety-critical domains—medical reasoning, legal analysis, mathematical proof verification, autonomous planning, and financial risk assessment—erroneous early exit could silently truncate reasoning chains, producing plausible but incorrect conclusions. The dual failure pattern is particularly concerning for deployment: oscillatory dynamics can cause tokens to appear converged at intermediate layers while their representations have not stabilized, creating a false sense of reliability that standard confidence metrics would not flag.

Our diagnostic framework enables practitioners to identify models and deployment regimes where early exit is unsafe: the compound error analysis (§5.1) quantifies the minimum per-token accuracy required for reliable question-level output, and the dual failure pattern (§4.3) highlights token populations most vulnerable to premature exit. We recommend that any deployment of per-token early exit on reasoning models include question-level accuracy validation, not only aggregate token-level metrics. We foresee no direct negative societal impacts from this diagnostic analysis.

L Compute Resources

All experiments were conducted on a single machine equipped with $4 \times$ NVIDIA RTX PRO 6000 GPUs (96 GB VRAM each). The primary cost is forward-pass inference through frozen pretrained models to generate per-token hidden-state checkpoints across multiple layers; lightweight probe training (tuned lens affine transforms, learned MLP gate) adds negligible overhead (< 1 GPU-hour total). The largest single experiment (R1-Qwen H1 profiling, 1,114 questions $\times$ 36 layers) required ~ 8 GPU-hours; the Phi-4-reasoning 14B experiments ($n=180$, 40 layers, 16K-token traces) required ~ 12 GPU-hours each. Total compute across all experiments (including failed runs and pilots) is estimated at < 200 single-GPU-hours on RTX PRO 6000 hardware. No multi-GPU parallelism was used; each experiment occupied a single GPU. Software environment: Python 3.12.3, PyTorch 2.10.0+cu128, Transformers 4.57.6, NumPy 2.2.6, SciPy 1.17.1.

Table 23: Per-type proxy Spearman ρ for R1-Llama. Reasoning keywords show strong *negative* ρ at L24, masked by the positive overall correlation.

Layer	Numerical	Special	Reasoning KW	Other
L16	0.045	0.168	−0.052	0.146
L20	0.088	0.182	−0.035	0.209
L24	0.307	0.174	− 0.515	0.372
L28	0.132	0.069	−0.265	0.162

Table 24: Oracle savings on R1-Qwen3-8B ($n=1,490,736$ tokens, 9 exit layers from a 36-layer model). Per-type breakdown at the 95% and 99% accuracy constraints.

Token Role	95% Accuracy		99% Accuracy	
	Savings	Avg Exit Layer	Savings	Avg Exit Layer
Overall	19.8%	L28.9	15.8%	L30.3
Numerical	19.4%	L29.0	15.4%	L30.5
Special	18.4%	L29.4	14.4%	L30.8
Reasoning KW	17.9%	L29.6	13.9%	L31.0
Other	20.5%	L28.6	16.5%	L30.1
Nested bound	19.2%	L29.1	15.2%	L30.5

M Dataset and Model Licenses

LLM usage declaration. Claude (Anthropic) [Anthropic, 2025] was used to assist with code writing and manuscript editing (grammar, formatting). No LLMs were used for data generation, experimental evaluation, or any other component of the research methodology—all experimental results derive from deterministic forward passes through the studied models with fixed random seeds.

N Sample Sizes

O Exit Accuracy Details

In Qwen3-8B (non-reasoning), exit accuracy varies widely by type (34.2%–60.5%, a 26.3pp range), with reasoning keywords the most accessible. Reasoning training compresses this distribution to a narrow 5.2pp band (28.5%–33.7%) compressing the most accessible types: reasoning keywords show the largest drop (−28.8pp), while numerical tokens—already the weakest in the non-reasoning model—drop only 5.7pp.

Depth profile and exit heatmap. R1-Qwen exit accuracy accumulates gradually: 0.5% at L8, 1.5% at L16, 3.7% at L20, 17.4% at L24, 32.4% at L28, 61.6% at L32. Llama’s profile is front-loaded by comparison: 25.9% at L20, 55.9% at L24 (+30pp in 12.5% of depth). This back-loaded computation in reasoning models means the layers most amenable to skipping contribute the least to prediction quality.

P Exit Strategy Details

Probe-based learned gate. We use the logistic regression probes from §5.1 as an exit gate: at layers L12–L28, a token exits at the shallowest layer where $P(\text{correct}) \geq \tau$. Table 31 shows the full threshold sweep on the 100-question test set (283,889 tokens).

The Pareto frontier is degenerate: no operating point achieves both non-trivial accuracy and non-trivial savings. At $\tau=0.95$, token accuracy reaches 98.9% but question accuracy remains 0%—a single misclassified token among $\sim 2,800$ per question suffices to corrupt the answer. The probe gate with $\text{AUC}=0.87$ achieves *lower* savings than cosine-similarity (0.054% vs. 0.34%) at its only non-zero question accuracy operating point, despite having a strictly superior exit signal. This is consistent with the interpretation that the bottleneck is not exit signal quality: under raw logit lens projection, too

Table 25: Probe AUC comparison: R1-Qwen (500q) vs. Qwen3-8B (non-reasoning) (100q). The non-reasoning model achieves uniformly higher AUC at all layers. Per-type breakdown at L28 shows reasoning keywords account for the largest gap (+0.169).

	R1-Qwen	Qwen3-8B (non-reasoning)	Δ (non-reas. - R1)
<i>Overall AUC by layer</i>			
L12	0.872	0.910	+0.038
L16	0.857	0.900	+0.043
L20	0.836	0.857	+0.021
L24	0.779	0.851	+0.072
L28	0.778	0.849	+0.071
<i>Per-type AUC at L28</i>			
Numerical	0.879	0.866	-0.013
Special	0.745	0.823	+0.078
Reasoning keyword	0.666	0.835	+0.169
Other	0.763	0.847	+0.084
<i>Exit accuracy at L28 (for context)</i>			
Overall	32.4%	47.7%	+15.3pp

Table 26: HumanEval results split by truncation status ($n=164$: 116 non-truncated, 48 truncated). Δ = truncated - non-truncated.

Metric	Non-truncated	Truncated	Δ	Significant?
H1: Overall non-mono	16.9%	12.2%	-4.7pp	No
H1: Numerical non-mono	36.4%	29.7%	-6.7pp	No
H1: Reasoning KW non-mono	4.4%	1.7%	-2.7pp	No
H2: Exit acc L28	37.8%	36.6%	-1.2pp	No
H2: Exit acc L32	64.0%	62.2%	-1.8pp	No
Mean tokens/question	2,238	8,192	—	—

few tokens produce correct predictions at early layers, regardless of how well one can *identify* them. A tuned lens analysis (Appendix T) reveals this reflects projection quality rather than information absence in the representations.

Learned gating (V2). Building on the V1 proof of concept (ExitGate with dropout=0.1, using raw logit-lens projection, which achieved $Q_{\text{Acc}}=53\%$ at $\sim 0.8\%$ savings), we implement four gating variants using tuned lens confidence (Table 32; evaluated on MATH-500; V2a: 5-fold stratified CV; V2b-V2d: 400/100 train/test, single seed). V2a (a learned MLP gate, MLP(13 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 1, no dropout) over 9 confidence features and 4 type indicators, at $\gamma=0.99$) achieves $Q_{\text{Acc}}=63.0\% \pm 5.2\%$ (nested CV: outer fold trains tuned lens, inner fold trains gate; 95% CI [56.5%, 69.5%]) at 6.28% savings—exceeding our pass criterion ($Q_{\text{Acc}} \geq 50\%$ at savings $\geq 5\%$) and substantially outperforming CALM baselines (0% softmax; 29.4%/11.4% tuned lens at $\tau=0.99$). V2b (per-type threshold grid search applied to V2a’s gate output) trades accuracy for aggressive savings (41.06% savings but $Q_{\text{Acc}}=13\%$, 75% early exit). V2c (per-type MLPs, one MLP(9 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 1) per token type, trained on confidence features) matches V2a’s accuracy ($Q_{\text{Acc}}=64\%$) with slightly lower savings (18.14%), indicating that explicit per-type structure provides no measurable advantage over a single MLP that takes type indicators as input features. V2d (logistic regression with StandardScaler + L2 on the same 13 features, at $\gamma=0.99$) reaches $Q_{\text{Acc}}=91\%$ at 3.75% savings; this near-full-depth operating point approximates the accuracy ceiling on the same 100-question test set, and the gap between V2a (63.0%) and the near-full-depth $\approx 91\%$ confirms compound error as the binding deployment constraint (Table 50).

Sequential split and cross-validation. To prevent information leakage from tuned lens training into gate evaluation, we adopt **nested cross-validation**: the outer fold trains per-layer affine transforms (tuned lens) on 400 questions, while the inner fold trains and evaluates the gate on the held-out 100 questions. This yields $Q_{\text{Acc}} = 63.0\% \pm 5.2\%$ (95% CI [56.5%, 69.5%]), lower than the non-nested estimate (68.2%) due to eliminating shared tuned-lens information between training and test sets.

Table 27: Token-level vs. question-level non-monotonicity rates. Token-level rates are computed over the full sample for each model; question-level rates are from the balanced 4×100 subsample used in the ANOVA (§C). Sample sizes differ because the token-level analysis uses all available data per model.

Model	Token-level (n_q)	Question-level (n_q)	Tokens/question
R1-Qwen3-8B	20.8% (1,114)	12.4% (100)	$\sim 2,980$
R1-Llama	19.3% (100)	14.3% (100)	~ 620
Qwen3-8B	11.6% (100)	10.6% (100)	$\sim 3,847$
Llama-3.1-8B	8.5% (300)	4.4% (100)	~ 295

Table 28: Licenses for datasets and models used in this study.

Asset	License
GSM8K [Cobbe et al., 2021]	MIT
MATH-500 [Hendrycks et al., 2021]	MIT
HumanEval [Chen et al., 2021]	MIT
AIME 2024	© MAA; fair use
DeepSeek-R1 distillations	MIT
Qwen2.5-1.5B-Instruct	Apache 2.0
Qwen3-8B	Apache 2.0
Llama-3.1-8B-Instruct	Llama 3.1 Community
Phi-4 (base)	MIT
Phi-4-reasoning-14B	MIT

Feature importance analysis. The V2d logistic-regression coefficients reveal that tuned lens confidence features dominate: margin (+0.976), top10_mass (−0.636), and log_max_prob (+0.576) account for nearly all predictive power ($|\text{coef}| > 0.5$). Type indicator features contribute minimally ($|\text{coef}| < 0.02$), confirming that the gating mechanism’s success derives primarily from tuned lens confidence signals rather than type identity per se. This is consistent with the V2a–V2c parity: both achieve ~ 63 – 64% Q_{Acc} despite V2c using explicit per-type structure, because the underlying exit signal—projection confidence at intermediate layers—is largely type-invariant once conditioned on margin and mass.

Table 33 shows the per-type breakdown for V2a ($\gamma=0.99$, original 400/100 split; 5-fold CV aggregation not performed per type). Although type features have negligible direct weight in the exit decision (see feature importance above), the gate’s confidence-based thresholds produce differentiated exit behaviour across token roles: numerical tokens exit at 68.5% while reasoning keywords exit at only 37.88%, reflecting the convergence dynamics documented in §4.2–§4.1. This confirms the value of token role as a diagnostic lens for understanding *why* the gate behaves differently across positions, rather than as a predictive feature driving exit decisions. At $\gamma=0.99$ all categories achieve $\geq 99.3\%$ token accuracy (other: 99.30%; special: 99.55%; numerical: 99.31%; reasoning keyword: 100%), with numerical tokens contributing the largest savings (25.07%) due to their high exit rate. V2a uses nested cross-validation (95% CI [56.5%, 69.5%]); V2b–V2d are evaluated on a single 400/100 train/test split ($n=100$, single seed). All variants use a single model (R1-Qwen3-8B) and dataset (MATH-500); cross-architecture transfer is validated in §5.2 and Appendix V.

826 P.1 Gate Architecture Ablation

827 To validate our choice of a 2-layer MLP gate, we compare four architectures on all three models using
828 pre-extracted features from distilled architectures (5-fold stratified CV, $\gamma=0.99$, seed 42). Table 36
829 reports Q_{Acc} / savings and validation loss.

830 Logistic regression is maximally conservative ($Q_{\text{Acc}} > 90\%$ but $\leq 4.6\%$ savings). The 2-layer MLP
831 achieves the best balance: highest Q_{Acc} on R1-Qwen (89.2%) and Phi-4 (81.6%) with the lowest
832 cross-fold variance. Adding a third layer provides no improvement and doubles variance on R1-Qwen
833 (std 8.0% vs. 1.9%), consistent with the 13-dimensional input not benefiting from additional capacity.

Table 29: Sample sizes (number of questions) by model and analysis.

Analysis	R1-Qwen	R1-Llama	Qwen3-8B (non-reasoning)	Llama-3.1
Convergence profiling (H1)	1,114	100	100	300
Exit accuracy (H2)	500	100	100	200
End-to-end simulation	500	100	100	200
OOD (HumanEval)	164	—	—	—

Table 30: Per-type exit accuracy at L28 for the Qwen architecture pair (R1: 500q, non-reasoning: 100q).

Token role	R1-Qwen	Qwen3-8B (non-reasoning)	Δ
Numerical	28.5%	34.2%	−5.7pp
Reasoning keyword	31.7%	60.5%	−28.8pp
Special	31.7%	49.4%	−17.7pp
Other	33.7%	49.8%	−16.1pp

P2 Exit Layer Grid Sensitivity

Table 37 compares exit layer grids on two models (R1-Qwen3-8B and R1-Llama-8B, MATH-500, nested CV, $\gamma=0.99$). The 7-layer grid omits the extreme exit points; both models show directionally consistent trends (fewer exit layers $\rightarrow$ higher Q_{Acc} , lower savings), though the effect size differs (+12.8 pp on R1-Qwen vs. +7.4 pp on R1-Llama) and 95% confidence intervals overlap for R1-Llama, so we describe the cross-model pattern as directionally consistent rather than statistically confirmed.

Q OOD Generalization

We evaluate R1-Qwen3-8B on HumanEval [Chen et al., 2021]—164 code generation problems requiring algorithmic reasoning.

Exit accuracy on HumanEval closely mirrors math results: L32 accuracy is 63.0% vs. 61.6% (+1.4pp), and the proxy remains uninformative (max $\rho = 0.19$; Table 38). Non-monotonicity is lower (14.0%) but still far above the non-reasoning baseline ($\sim 8.5\%$). Per-type rankings shift across domains (Appendix Q), reinforcing that no universal per-type exit policy exists.

OOD per-type visualization.

R Cross-Difficulty and Cross-Domain Validation

To test whether our findings extend to substantially harder mathematical reasoning, we profile 30 AIME 2024 competition problems using DeepSeek-R1-0528-Qwen3-8B (max 16,384 new tokens; 22/30 traces truncated at this limit, averaging $\sim 15,575$ tokens per question). Table 39 summarizes the results alongside the main GSM8K findings.

All five metrics are directionally consistent. The slightly elevated non-monotonicity (23.2% vs. 20.8%) and comparable exit accuracy confirm that harder problems do not qualitatively alter the convergence dynamics documented in §4–5. The high truncation rate (73%) suggests that longer reasoning traces, if anything, amplify the challenges for early exit.

Cross-difficulty tuned lens oracle (MATH-500). We evaluate the GSM8K-trained tuned lens affine transforms on MATH-500 (500 questions, R1-Qwen3-8B) to test cross-difficulty generalization. **Methodology note:** both the GSM8K training and MATH-500 evaluation use dataset reference solutions under teacher forcing (GSM8K: ~ 168 tokens/question; MATH-500: ~ 304 tokens/question), not model-generated reasoning traces. The distribution shift thus reflects problem difficulty and solution style rather than the trace-length differences observed in model-generated output (Table 18).

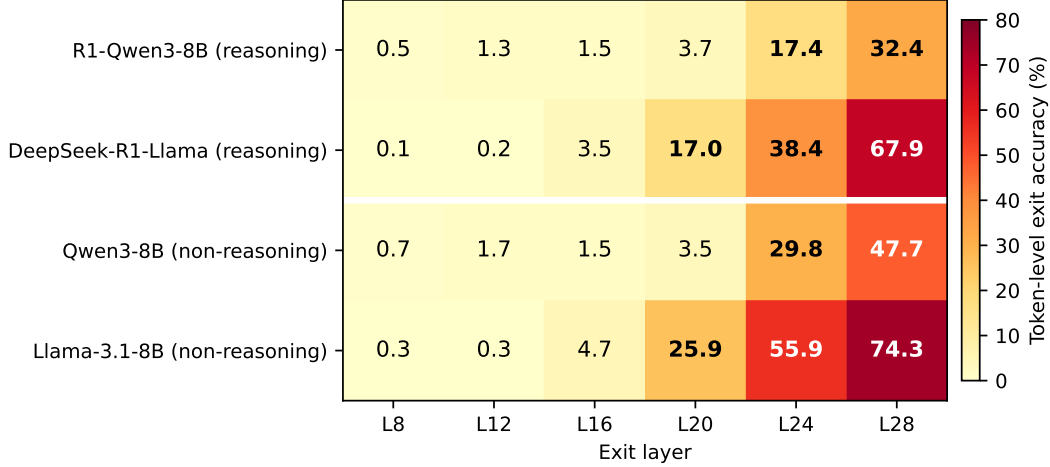


Figure 5: Token-level exit accuracy across the 2×2 design (4 models $\times$ 6 checkpoint layers). Reasoning models (top) show uniformly lower accuracy than non-reasoning models (bottom), with the sharp divergence concentrating in the final 25% of layers. L28 = 78% depth for Qwen, 88% for Llama.

Table 31: Probe gate threshold sweep. Q_{Acc} is 0% at all thresholds ≤ 0.95 ; the first non-zero operating point ($\tau=0.99$, 23% Q_{Acc}) delivers less savings than cosine-similarity (0.054% vs. 0.34%).

Threshold τ	Token Acc (%)	Question Acc (%)	Savings (%)
0.50	47.1	0	30.4
0.70	75.4	0	15.6
0.80	87.5	0	8.8
0.90	96.3	0	3.1
0.95	98.9	0	0.94
0.99	99.9	23	0.054

864 The gap structure (raw $\ll$ tuned) is preserved across domains: tuned lens consistently outperforms
865 raw projection. However, the tuned oracle drops from 58.2% to 35.4% (-22.8pp), with early layers
866 most affected (L8: -25.0pp) and late layers relatively preserved (L34: -9.4pp). This degradation
867 is expected: the affine transforms are trained on GSM8K hidden states and partially overfit to that
868 distribution. The raw oracle’s stability (12.6% vs. 12.2%) confirms that the degradation is in the
869 learned projection, not in the underlying representation quality. Domain-specific tuned lens training
870 would likely recover much of the gap.

871 S Answer Segment Analysis

872 We segment each reasoning trace into three regions—*reasoning* (before `</think>`), *answer* (after
873 `</think>` excluding `\boxed{\}`), and *boxed* (tokens inside `\boxed{\}`)—and compute per-token exit
874 accuracy within each segment (Table 41). Data: R1-Qwen3-8B, 500 GSM8K questions.

875 Two patterns emerge: (1) at lower/middle layers (L4–L20), exit accuracy is nearly uniform across
876 segments; (2) at upper layers, boxed tokens converge fastest (L32: 78.3%), followed by answer tokens
877 (65.1%), with reasoning tokens slowest (61.4%). The degradation curve is stark: R1-Qwen3-8B
878 achieves 84.3% GSM8K accuracy at full depth (L35, 497 questions with `</think>`), but evaluating
879 only the boxed-answer subset ($n=200$), answer correctness collapses to 1.6% at L32 and 0% at
880 L28. Despite boxed tokens having the highest per-token accuracy at L32, compound error across the
881 multi-token answer sequence is the primary bottleneck.

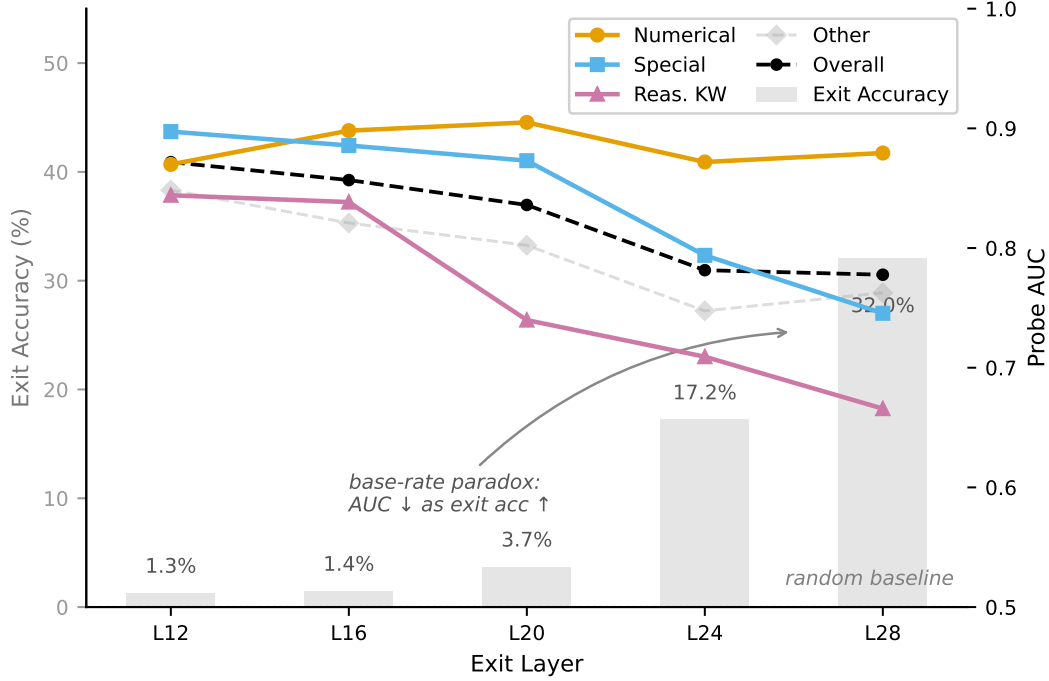


Figure 6: Probe AUC (lines) and token-level exit accuracy (background bars) across layers for R1-Qwen3-8B. High AUC at early layers (0.87 at L12) contrasts with near-zero exit accuracy (1.3%), illustrating the base-rate paradox: correct-exit tokens are identifiable but scarce. Per-type AUC shows reasoning keywords are hardest to discriminate (0.67 at L28).

Table 32: V2 learned gating variants (R1-Qwen3-8B, MATH-500 (400 train / 100 test), 9-layer exit grid, tuned lens projection). Pass criterion: $Q_{\text{Acc}} \geq 50\%$ at savings $\geq 5\%$. V2a: nested CV (95% CI [56.5%, 69.5%]); V2b–V2d: $n=100$, single seed.

Variant	Strategy	Q_{Acc}	Savings	Early Exit
V2a	Learned MLP gate ($\gamma=0.99$)	63.0%	6.28%	—
V2b	Layer gating (per-type threshold search)	13%	41.06%	75%
V2c	Per-type MLP gate	64%	18.14%	52%
V2d	Logistic regression (StandardScaler + L2)	91% [†]	3.75%	33%

[†]Binomial 95% CI: [83.6%, 95.8%], $n=100$.

T Tuned Lens Comparison

Raw logit lens projection (applying the LM head to intermediate hidden states without correction) systematically underestimates layer capability [Belrose et al., 2024]. We train per-layer affine transforms T_ℓ on 400 GSM8K train questions (separate GSM8K tuned lens instance, distinct from the MATH-500 nested CV lens) (KL divergence to final-layer logits) and evaluate on 100 held-out test questions. Training uses Adam ($\text{lr}=10^{-3}$), token batch size 4,096, up to 10 epochs with early stopping (patience 3, $\delta=10^{-5}$), and identity initialization ($W=I$, $b=0$). The per-layer affine transform (d^2+d parameters: a $d \times d$ weight matrix plus bias; $d=4096$ for 8B models) adds overhead equivalent to $\sim 7\%$ of one transformer layer—far less than the ~ 4 full layers saved at the 11.4% operating point. Table 42 compares raw and tuned exit accuracy for both R1-Qwen3-8B and Qwen3-8B (non-reasoning).

Tuned lens dramatically improves absolute exit accuracy for *both* model types: R1 gains +54pp at L24 and +48pp at L28; the non-reasoning model gains +58pp and +51pp at the same layers. Crucially, the reasoning model’s deficit relative to the non-reasoning model **persists** under tuned lens (4–6pp at L24–L32), confirming that the exit accuracy deficit documented in §4.2 reflects a genuine property of reasoning representations, not a systematic bias of the logit lens projection method.

Table 33: V2a per-type breakdown ($\gamma=0.99$). The gate automatically learns differentiated exit strategies: numerical tokens exit aggressively while reasoning keywords are retained. Per-type exit rates are from the original 400/100 partition; 5-fold stratified CV is applied to question-level Q_{Acc} only.

Token Role	Exit Rate	Token Acc
Numerical	68.50%	99.31%
Reasoning keyword	37.88%	100.00%
Other	48.18%	99.30%
Special	55.70%	99.55%

Table 34: V2a MLP gate nested cross-validation results on MATH-500 (R1-Qwen3-8B). Outer fold trains tuned lens on 400 questions; inner fold trains and evaluates gate on 100 held-out questions. Mean $Q_{\text{Acc}} = 63.0\% \pm 5.2\%$ (95% CI: [56.5%, 69.5%]).

Model	Q_{Acc} (%)	Savings (%)
R1-Qwen3-8B	63.0 ± 5.2	6.28 ± 0.63
R1-Llama-8B	55.0 ± 8.9	9.41 ± 0.62
Phi-4-reasoning-14B	48.0 ± 7.3	12.70 ± 0.71

Oracle ceiling under tuned lens. Applying the tuned lens per-token correctness to the oracle framework (§5.2) on 500 GSM8K test questions, the lossless oracle ceiling rises from 12.6% to **58.2%** savings (+45.6pp). At L8, 45.0% of tokens can exit under tuned lens (vs. 0.5% raw), and oracle coverage reaches 93.5% (vs. 77.8% raw). This suggests that projection quality—rather than information absence—is the dominant factor: intermediate layers contain sufficient information for correct predictions when accessed through learned transforms, but raw projection fails to extract it.

Note on evaluation set. The tuned lens exit accuracy evaluation (Table 42, 100q columns) uses 100 GSM8K test questions disjoint from the 400-question training set, whereas the 500q columns and the main exit accuracy results in §4.2 use 500 questions. The 100-question evaluation yields lower raw accuracy values (e.g., L28: 24.5% here vs. 32.4% in §4.2) due to the different hidden state extraction point required by tuned lens (pre-layer-norm residual stream vs. post-FFN in the main experiments); the 500q columns confirm this offset is consistent across sample sizes. The v3 oracle (9-layer grid, 500 test questions) yields 58.2% tuned lossless savings.

Extraction point control. We decompose the tuned lens improvement into three components: (1) extraction point (pre-layer-norm vs. post-FFN) contributes negatively (−0.1 to −6.6pp), (2) applying the final RMSNorm to pre-LN states explains <10% of the gap, and (3) the learned affine transform accounts for >90% of the +43–58pp improvement, confirming that the benefit is genuine representation alignment, not an extraction artifact.

Phi-4 14B tuned lens. Extending the tuned lens analysis to Phi-4-reasoning (14B, 40 layers; 400 train / 100 test questions) confirms cross-architecture consistency. Table 43 reports per-layer exit accuracy; tuned lens improves accuracy by +46.8pp on average (range: +11.9–68.4pp), closely matching R1-Qwen (~+47pp). Peak tuned accuracy reaches 91.8% at L37 rather than L38—a non-monotonic pattern paralleling the raw exit accuracy finding (§4.1). The oracle ceiling rises from 27.3% (raw) to 76.6% (tuned, +49.3pp); at 95% token accuracy, savings increase from 32.3% to 81.2%, paralleling R1-Qwen (+45.6pp) and R1-Llama (+36.9pp). However, CALM-style adaptive exit ($\tau=0.90$) achieves only 2.8% Q_{Acc} at 16% savings, confirming the oracle-to-deployment transfer gap widens at 14B scale.

A per-type reversal emerges under tuned lens: raw projection favors content-bearing tokens (reasoning keywords 29.8–84.7%, numerical 7.2–87.5% across layers), while tuned lens favors structural tokens (other 76.6–91.5%, special 80.4–95.5%). Reasoning keyword and numerical tokens recover to only ~54–86% under tuned lens at early layers, vs. >75% for other/special. This suggests content-bearing tokens encode information non-linearly in intermediate layers, limiting affine transform recovery.

Table 35: V2a gate Q_{Acc} by MATH-500 difficulty level (5-fold CV aggregate).

Difficulty	Q_{Acc} (%)
Level 1	64.5
Level 2	70.0
Level 3	67.6
Level 4	75.6
Level 5	61.1

Table 36: Gate architecture comparison across three reasoning models ($\gamma=0.99$, 5-fold CV, pre-extracted features). Numbers are from a controlled comparison holding feature extraction constant; absolute values differ from nested CV results in §5.2. The 2-layer MLP(64→32) achieves the best Q_{Acc} -variance tradeoff; deeper gates show no benefit.

Gate	R1-Qwen Q_{Acc} (%)	R1-Llama Q_{Acc} (%)	Phi-4 Q_{Acc} (%)
LogReg	98.0±1.4 / 0.1%	94.2±2.0 / 4.6%	90.8±0.8 / 3.5%
1-layer MLP(64)	86.0±4.7 / 4.9%	71.4±4.6 / 9.1%	83.0±4.3 / 5.6%
2-layer MLP(64→32)	89.2±1.9 / 4.3%	65.8±4.2 / 9.4%	81.6±2.5 / 6.6%
3-layer MLP(64→32→16)	86.0±8.0 / 4.7%	62.2±4.9 / 10.0%	82.8±5.9 / 7.2%

T.1 Computational Overhead Analysis

U Native Thinking Mode Full Results

Table 45 reports the full Qwen3-8B native thinking mode analysis ($n=499$ with `</think>`; 1 of 500 questions excluded). This model uses Qwen3-8B’s built-in thinking mode (no distillation), providing a controlled comparison against the distilled R1-Qwen3-8B on the same architecture.

Unpaired token-level comparison: thinking vs. base. A two-proportion z -test on token-level NM rates comparing Qwen3-8B thinking mode (967K tokens, NM=9.11%) against Qwen3-8B base (385K tokens, NM=11.58%) yields $z=-43.48$ ($p\approx 0$, $\chi^2=1,890$): statistically significant but small effect size (Cohen’s $h=0.08$), consistent with §4.4 that oscillatory dynamics are distillation-specific rather than reasoning-specific. The large z -statistic reflects the sample size (1.35M tokens combined); the actual difference is 2.47pp. Per-type hierarchy is preserved in thinking mode: numerical (16.0%) > reasoning keyword (11.5%) > special (8.3%) > other (7.3%). This is an unpaired comparison across different question sets (500 vs. 100 questions, not matched per-question); a within-question paired design would strengthen this finding.

V Scale Analysis

Our 8B findings raise a natural question: do the non-monotonicity gap (H1) and exit-accuracy deficit (H2) persist at other scales? We extend the analysis to 1.5B ($n=200$), 14B ($n=180$), and 32B ($n=500$), revealing a scale-dependent *reversal* at 1.5B that constrains the generality of the 8B conclusions, while CALM failure persists at 32B.

1.5B reversal. Table 46 summarizes the cross-scale comparison. At 1.5B, both H1 and H2 reverse direction relative to the 8B pattern: the non-reasoning baseline (Qwen2.5-1.5B-Instruct) exhibits *higher* non-monotonicity than the reasoning model (R1-Distill-Qwen-1.5B), and the reasoning model achieves *higher* exit accuracy at all comparable checkpoints. Question-level bootstrap confidence intervals (1,000 iterations) confirm statistical robustness: the 95% CIs are non-overlapping (R1-Distill: [12.2, 16.3]%; Qwen2.5: [27.7, 31.8]%), with an 11.4 pp gap between the nearest edges.

Layer-by-layer exit accuracy (H2). The reversal is not confined to a single checkpoint. Table 47 reports exit accuracy across layers for the 1.5B pair, alongside the 14B Phi-4 base trajectory. At 1.5B, R1-Distill leads at every checkpoint from L8 through L24 (e.g., L16: 10.9% vs. 5.3%; L20: 41.6% vs. 32.5%), the opposite of the 8B pattern where reasoning consistently trails non-reasoning at early

Table 37: Exit layer grid sensitivity ($\gamma=0.99$, 5-fold nested CV, MATH-500). Both models show the same directional trend; effect size is architecture-dependent.

Model	Grid	Q_{Acc}	Savings	ΔQ_{Acc}
R1-Qwen (36L)	9-layer (4,8,...,34)	63.0% $\pm$ 5.2%	6.28% $\pm$ 0.63%	—
	7-layer (8,12,...,32)	75.8% $\pm$ 2.8%	4.08% $\pm$ 0.29%	+12.8 pp
R1-Llama (32L)	9-layer (4,8,...,30)	55.0% $\pm$ 8.9%	9.41% $\pm$ 0.62%	—
	7-layer (4,8,...,28)	62.4% $\pm$ 7.1%	8.36% $\pm$ 0.49%	+7.4 pp

Table 38: OOD generalization: HumanEval (code) vs. GSM8K/MATH (math) on R1-Qwen3-8B ($\theta = 0.03$).

Metric	Math (in-dist)	HumanEval (OOD)	Δ
Non-monotonicity (overall)	20.8%	14.0%	−6.8pp
Exit accuracy L28 ($d=0.78$)	32.4%	37.1%	+4.7pp
Exit accuracy L32 ($d=0.89$)	61.6%	63.0%	+1.4pp
Proxy ρ (max across layers)	<0.15	0.19	—

exits. The reversal is most pronounced at early layers (L8: 5.4% vs. 2.2%, a $2.5\times$ ratio) and narrows toward the final layer.

Interpretation: baseline instability dominates at 1.5B. At 8B and above, reasoning training is the dominant source of convergence difficulty—reasoning models show higher NM and lower exit accuracy than their non-reasoning counterparts. At 1.5B with distilled models, this relationship reverses: the non-reasoning baseline (Qwen2.5-1.5B-Instruct) already exhibits substantial representation instability (NM=29.7%), and distillation-based reasoning training does not add to this baseline difficulty. Both models retain substantial mathematical capability (R1-Distill: 81.5%, Qwen2.5: 69.0% on GSM8K), so the reversal is not an artifact of model incapacity.

Caveats and scope. Four factors limit the strength of conclusions drawn from the 1.5B reversal:

- Special-token inflation.** Aggregate NM includes punctuation and LaTeX formatting tokens ($\backslash$, $.$, $=$, $\{$, $\}$ etc.), which exhibit disproportionately high non-monotonicity. In Qwen2.5-1.5B-Instruct, the LaTeX backslash token alone reaches NM=97.8%; special tokens overall show NM=54.0%. Excluding special tokens, the gap narrows from 15.6pp to 9.9pp (R1-Distill 12.7% vs. Qwen2.5 22.6%), but the reversal persists.
- Non-identical base architectures.** R1-Distill-Qwen-1.5B is distilled from DeepSeek-R1 into a Qwen2-1.5B backbone, while Qwen2.5-1.5B-Instruct is based on Qwen2.5-1.5B. These differ in pretraining data, vocabulary, and architecture details, so the comparison is not a controlled ablation isolating reasoning training alone.
- Two-point extrapolation.** We observe the expected pattern at 8B and a reversal at 1.5B—only two data points. We use “reversal” rather than “phase transition” to avoid implying a well-characterized boundary; the crossover scale is unknown and may depend on training methodology (distillation vs. native reasoning).
- Dip-layer distribution differs.** Non-monotonic dips in R1-Distill-1.5B concentrate in early and middle layers, while those in Qwen2.5-1.5B-Instruct concentrate in early and deep layers. The reversed aggregate NM thus reflects different *patterns* of instability, not merely different magnitudes—the two models may fail for qualitatively different reasons.

14B: Phi-4 reasoning–base comparison. Phi-4-reasoning (14B, 40 layers, $n=180$; 16K token budget) shows NM=25.5% (trimmed), more than double the 11.6% base rate ($\Delta=+13.98$ pp, 95% CI [12.17, 14.62] pp). The base model ramps from 18.6% (L8) to 91.2% (L36); the reasoning model starts lower (10.3% at L8) but surpasses it by L36 (96.7%). Only 9/180 traces are untruncated (1,050–3,442 tokens); the naïve untruncated-vs-truncated $\Delta=12.4$ pp is confounded by difficulty (shorter traces = easier questions = lower NM). Position-controlled comparison—matched-length prefixes: $\Delta\approx 1.3$ pp; tail-vs-head segments: $\Delta=2.96$ pp—and a cross-model proxy (R1-Qwen-8B,

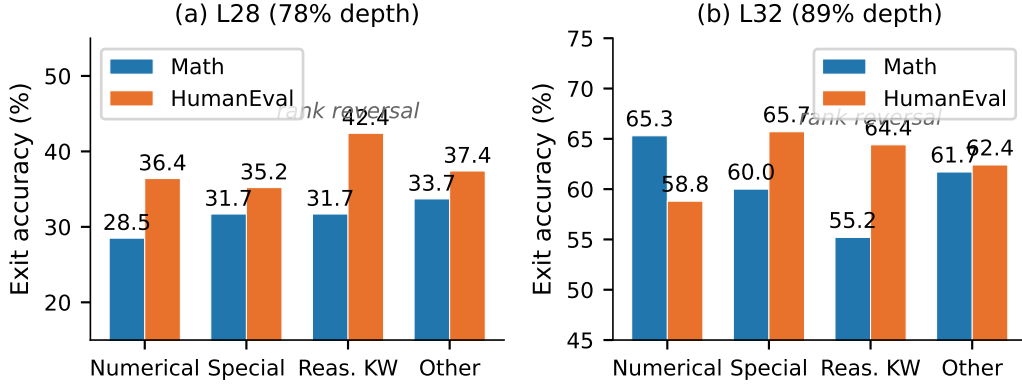


Figure 7: OOD generalization to HumanEval code reasoning. **Left (a)**: Per-type exit accuracy at L28 (78% depth). **Right (b)**: Per-type exit accuracy at L32 (89% depth). Ranking reversals between Math and HumanEval are annotated. Overall accuracy closely mirrors math results (+1.4pp at L32).

Table 39: AIME 2024 vs. GSM8K: core metrics remain directionally consistent on harder problems ($n=30$, 73% truncated).

Metric	AIME (30q)	GSM8K (500q)
H1 non-mono ($\theta=0.03$)	23.2%	20.8%
Numerical	48.4%	42.9%
H2 exit acc (L28)	30.6%	32.4%
H2 exit acc (L32)	63.6%	61.6%
Proxy ρ (L28)	0.091	0.050

992 $n=27$; $|\Delta| < 0.4$ pp at all truncation points) bound the truncation artifact at ≤ 3 pp, small relative to
993 the 13.98 pp gap.

994 **Phi-4 penultimate layer effect.** The Phi-4 base 14B model exhibits a pronounced representation
995 shift at the penultimate layer transition (L38→L39): 86.4% of tokens show a cosine-similarity dip at
996 this boundary. Including this layer, the running-max NM rate reaches 70.0%—an order of magnitude
997 higher than the trimmed rate of 11.6% that excludes the final column. This penultimate-layer effect
998 is architecture-specific (Phi3ForCausalLM, 40 layers) and does not appear in the 8B Qwen/Llama
999 models we study. All Phi-4 NM rates reported in Table 46 use the trimmed definition to ensure
1000 cross-scale comparability; the trimmed rate of 11.6% is consistent with other non-reasoning baselines
1001 (Qwen3-8B: 11.6%, Llama-3.1-8B: 8.5%).

1002 **32B: QwQ-32B CALM diagnostic.** QwQ-32B [Qwen Team, 2024] (64 layers, $n=500$, 21 exit
1003 checkpoints) tests whether the CALM failure pattern persists at 32B scale. At $\tau=0.95$, CALM
1004 exits zero tokens: savings=0% and $Q_{\text{Acc}}=98.8\%$ (effectively full-depth inference). Lowering the
1005 threshold to $\tau=0.90$ immediately collapses to $Q_{\text{Acc}}=2.4\%$ at savings=0.2%, reproducing the all-
1006 or-nothing failure pattern observed at 8B–14B. No threshold between 0.90 and 0.95 yields a viable
1007 accuracy–savings trade-off. This confirms that the Reasoning Early Exit Gap is not a small-model
1008 artifact: standard CALM criteria fail on 32B reasoning models just as they do at 8B.

1009 **32B: QwQ-32B V2a gate.** Despite CALM failure, the V2a gate recovers meaningful exit perform-
1010 mance at 32B. Table 48 reports per-threshold results on MATH-500 ($n=500$, 5-fold nested CV). At
1011 $\gamma=0.99$, the gate achieves $61.0\% \pm 2.4\%$ Q_{Acc} at $4.98\% \pm 0.44\%$ savings—comparable Q_{Acc} to the
1012 8B R1-Qwen (63.0%) but with lower savings. The reduced savings reflect exit grid sparsity: 9 exit
1013 layers span 64 total layers (one exit per ~ 7.1 layers), so the shallowest possible exit saves at most
1014 $\frac{62}{64}=96.9\%$, but the practical maximum exit layer (L62, 2 layers from the final) saves only 3.1% per
1015 token. Gate val_loss averages well below 0.50, confirming genuine learned boundaries.

Table 40: Cross-difficulty tuned lens oracle: GSM8K (in-distribution) vs. MATH-500 (OOD). Raw oracle is difficulty-insensitive; tuned oracle drops substantially.

Metric	GSM8K	MATH-500
Raw oracle (lossless)	12.6%	12.2%
Tuned oracle (lossless)	58.2%	35.4%
Δ (tuned – raw)	+45.6pp	+23.2pp
Tuned coverage	93.5%	85.1%
<i>Per-layer tuned accuracy</i>		
L8	45.0%	20.0%
L24	65.7%	37.1%
L32	82.4%	68.6%
L34	86.9%	77.4%

Table 41: Per-token exit accuracy (%) by trace segment and answer correctness by layer. Answer correctness at L4–L32 is evaluated on the boxed-answer subset ($n=200$ questions containing `\boxed{\}`); L35 reports full-depth GSM8K accuracy over all 500 questions.

Layer	Overall	Reasoning	Answer	Boxed	Ans. Correct
L20	3.7	3.8	3.4	0.5	0%
L24	17.4	17.4	18.0	16.0	0%
L28	32.4	32.3	33.5	27.4	0%
L32	61.6	61.4	65.1	78.3	1.6%
L35 (final)	100.0	100.0	100.0	100.0	84.3% [†]

[†]Full-depth GSM8K accuracy (497 of 500q with `</think>`); L4–L32 use boxed subset ($n=200$).

Methodology notes. QwQ-32B uses `bfloat16` with `float32` accumulation (vs. full `float32` at 8B–14B) due to 32B memory constraints, and `answer_length_tercile` stratification (vs. `difficulty_level`) because MATH-500 difficulty labels are unavailable for this model’s outputs. Neither difference affects the gate training procedure.

7B: R1-Distill-Qwen-7B V2a gate. R1-Distill-Qwen-7B (28 layers, $n=500$; oracle accuracy 56.6%) extends the V2a learned gate evaluation to a smaller distilled model. At $\gamma=0.99$, the gate preserves $88.2\%\pm 1.9\%$ Q_{Acc} but achieves only $4.48\%\pm 0.41\%$ savings; at $\gamma=0.95$, Q_{Acc} drops to $30.0\%\pm 4.6\%$ at 12.63% savings. Notably, gate validation loss remains at ≈ 0.50 (chance level across all 5 folds), indicating the MLP gate does not learn meaningful exit boundaries on distilled traces—the same failure pattern observed in the non-reasoning control (Qwen2.5-7B-Instruct, `val_loss` ≈ 0.50). The nominally higher Q_{Acc} (88.2% vs. R1-Qwen’s 63.0%) reflects the gate’s conservative non-exit strategy at $\gamma=0.99$ (savings 4.48% vs. 6.28%), not superior signal quality. The sharp Q_{Acc} drop from $\gamma=0.99$ to $\gamma=0.95$ confirms compound error as the binding constraint across model scales.

14B: Phi-4-reasoning V2a gate (cross-architecture). Phi-4-reasoning-14B (40 layers, $n=500$, GSM8K; oracle accuracy 64.8%) extends V2a gate evaluation to a non-Qwen architecture. Table 49 reports per-threshold results. At $\gamma=0.99$, the gate achieves $48.0\%\pm 7.3\%$ Q_{Acc} (95% CI [39.0%, 57.0%]) at $12.70\%\pm 0.71\%$ savings—gate signal transfers cross-architecture, though Q_{Acc} decreases substantially (48.0% vs. 63.0% on R1-Qwen) while savings are modestly higher (12.70% vs. 6.28%). Despite higher savings, the lower Q_{Acc} suggests the gate’s decision boundary, learned on R1-Qwen’s representations, produces noisier exit decisions on Phi-4’s distinct representation space. Gate validation loss averages ≈ 0.41 across folds (range 0.407–0.412), well below the ≈ 0.50 random-classifier level observed in R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct (non-reasoning)—confirming that the V2a gate learns genuine exit boundaries on Phi-4’s representations despite never training on this architecture. Since R1-Llama (8B, distilled from the same R1 teacher) also yields successful gate training, while the smaller R1-Distill-Qwen-7B (28 layers, same teacher) does not, capacity and/or architecture differences may contribute to whether learnable exit signals emerge, though the current evidence does not isolate the mechanism. The steep Q_{Acc} degradation at lower thresholds (28.2% at $\gamma=0.95$; 0% at $\gamma=0.50$) mirrors the compound-error pattern across all models.

Table 42: Raw vs. tuned lens exit accuracy (%). 100q columns use the tuned lens evaluation set (disjoint from training); 500q columns use the main experiment set. Tuned lens raises absolute accuracy by 35–54pp, but the reasoning–non-reasoning gap persists (4–6pp at L24–L32).

Layer	R1-Qwen3-8B				Qwen3-8B (non-reasoning)			
	100q		500q		100q		500q	
	Raw	Tuned	Raw	Tuned	Raw	Tuned	Raw	Tuned
L20	2.3	51.4	2.4	50.6	3.2	57.1	—	—
L24	12.0	66.3	11.7	65.7	14.2	72.3	—	—
L28	24.5	72.5	23.9	72.1	27.3	77.8	—	—
L32	47.3	82.9	46.9	82.3	53.5	87.0	—	—

Table 43: Raw vs. tuned lens exit accuracy (%) for Phi-4-reasoning (14B, 40 layers; 100 test questions). Peak tuned accuracy at L37, not L38, reflecting non-monotonic representation dynamics at near-final layers.

Layer	Raw	Tuned	Δ (pp)
L5	6.9	75.3	+68.4
L10	9.4	77.3	+67.8
L15	9.7	76.9	+67.2
L20	14.6	79.3	+64.7
L25	26.2	83.8	+57.7
L30	47.6	88.0	+40.4
L35	65.3	91.0	+25.7
L37	74.7	91.8	+17.0
L38	77.8	89.6	+11.9

8B: R1-Llama V2a gate (cross-architecture). R1-Llama-8B (32 layers, $n=500$, GSM8K; oracle accuracy 65.2%) provides a second cross-architecture V2a transfer point within the same 8B scale. At $\gamma=0.99$, the gate achieves $55.0\%\pm 8.9\%$ Q_{Acc} at $9.41\%\pm 0.62\%$ savings; Q_{Acc} is lower than R1-Qwen (63.0% $\pm 5.2\%$) but savings are comparable, confirming that gate accuracy—but not savings efficiency—is architecture-dependent. As with Phi-4, the gate adopts a conservative exit strategy on the transfer target, confirming that gate accuracy—but not savings efficiency—transfers across architectures without retraining.

W Trace-Length and Temporal Analysis

Figure 8 decomposes the compound error mechanism by trace length (500 GSM8K questions, R1-Qwen3-8B). Per-token exit fidelity at L34 remains stable across all length buckets ($\sim 99.3\text{--}99.9\%$ of tokens exceed $\cos \theta = 0.60$), yet question-level accuracy—requiring *every* token to be faithful—decays monotonically from 78.3% (100–500 tokens) to 0% ($>5,000$ tokens). The decay closely tracks the theoretical p^T curve: at $p=0.999$, a 5,000-token trace has $0.999^{5000} \approx 0.7\%$ probability of all tokens being correct. This is consistent with interpreting the near-zero question accuracy reported in §5.1–5.2 as a trace-length-dependent compound error effect rather than a reasoning-specific failure mode.

Compound error quantification. Table 50 quantifies the compound error model $Q_{\text{Acc}} = p^T$ under the independence assumption, where p is per-token exit accuracy and T is trace length. Even at $p=0.999$ (higher than any observed raw-projection accuracy), a 3,000-token trace retains only 5.0% question accuracy. The final rows show our observed operating points: CALM cosine-similarity at $\tau=0.97$ achieves $p=0.992$ but $Q_{\text{Acc}}=0\%$ over $\sim 3,000$ tokens, closely matching the theoretical prediction ($0.992^{3000} < 10^{-10}$). Tuned lens at $\tau=0.99$ achieves higher Q_{Acc} than p^T predicts (29.4% vs. $\approx 0\%$), because tuned projection corrects *correlated* errors—the independence assumption is conservative. The “Required p ” row shows the minimum per-token accuracy needed for $\geq 50\%$ question accuracy: at $T=3,000$, this requires $p \geq 99.977\%$ —far beyond any current method.

Temporal dynamics within reasoning traces. We analyze how non-monotonic convergence varies *within* traces by dividing each trace into positional quartiles ($n=1,466$ questions, $\theta=0.03$; the main

Table 44: Computational overhead breakdown for tuned-lens gating (R1-Qwen3-8B, 36 layers, 9-layer exit grid). Net savings account for probe and gate evaluation costs. All costs are analytical estimates based on per-layer FLOPs ratios, not runtime profiling measurements.

Component	Cost
Affine transform (per layer)	$\sim 7\%$ of one layer
Gate MLP evaluation (per layer)	$< 0.1\%$ of one layer
Avg. layers evaluated before exit	~ 5 of 9 candidates
Total probe overhead	$\sim 1\%$ of full inference
Gross token savings (V2a)	6.28%
Net token savings	$\sim 5.3\%$
ROI (net savings / overhead)	$\sim 5:1$

Table 45: Native thinking mode vs. distilled reasoning (Qwen3-8B architecture). NM = non-monotonicity ($\theta=0.03$); Conv = fraction of tokens with $\cos > 0.9$ at L34.

Metric	Thinking	R1-Qwen	Qwen3 (non-reas.)
<i>H1: Non-monotonicity (per-type)</i>			
Overall	9.1%	20.8%	11.6%
Numerical	16.0%	42.9%	15.3%
Special	8.3%	27.4%	18.9%
Reasoning KW	11.5%	4.2%	7.8%
Other	7.3%	11.8%	8.5%
<i>H2: Exit accuracy</i>			
L28	34.7%	32.4%	47.7%
L32	62.6%	61.6%	73.3%
<i>Temporal NM (answer segment)</i>			
Answer NM	10.0%	32.7%	—
<i>Difficulty (L34 token-weighted convergence by tercile)</i>			
Short traces	72.5%	57.9%	—
Long traces	85.2%	29.6%	—
Direction	<i>reversed</i>	<i>expected</i>	—
<i>Oracle savings (95% nested bound, 9-layer grid)</i>			
Savings	20.0%	19.2%	—
<i>Proxy correlation</i>			
ρ (cos-sim vs. exit)	0.109	—	—
Ground-truth accuracy	95.6%	84.3%	—

1070 text reports $n=1,411$ after filtering traces with fewer than 20 reasoning tokens). Table 51 reveals a
1071 U-shaped profile: elevated non-monotonicity at trace start (exploration phase), a minimum in the
1072 third quartile (convergent reasoning), and a sharp spike in the final quartile associated with answer
1073 production.

1074 The Q4 spike is explained by the answer segment: tokens after the `</think>` delimiter exhibit 32.7%
1075 non-monotonicity versus 15.1% in the reasoning body ($2.17\times$). The model’s hidden states become
1076 less stable when transitioning from reasoning to answer output, precisely where correctness matters
1077 most for question-level accuracy. Reasoning keywords show the strongest positional decay (4.6%
1078 $\rightarrow 1.4\%$, a $3.2\times$ reduction), indicating that logical connectives stabilize progressively as the trace
1079 deepens.

1080 **Trace-length stratification.** We stratify the 500-question R1-Qwen analysis by reasoning trace
1081 length. Table 52 shows that H1 non-monotonicity rates are stable across trace-length quintiles
1082 ($\rho=-0.14$, $p=0.002$), while question accuracy under oracle exit decays from 47% to 0%—consistent
1083 with compound error p^T as the dominant factor.

Table 46: Cross-scale comparison (running-max NM, $\theta=0.03$; $n=200$ for 1.5B/8B, $n=180$ for 14B). H1 = non-monotonicity rate; H2 = exit accuracy at the deepest shared checkpoint. “Direction” indicates whether reasoning > non-reasoning (H1) or reasoning < non-reasoning (H2), as predicted by the 8B findings; *reversed* = opposite.

Scale	H1 reasoning	H1 non-reas.	H1 dir.	H2 reasoning	H2 non-reas.
1.5B (28L, L24)	14.1% [12.2, 16.3]	29.7% [27.7, 31.8]	<i>reversed</i>	81.3%	79.2%
8B (36L, L28)	20.8%	11.6%	✓	32.4%	47.7%
14B (40L, L36)	25.5% [†]	11.6% [†]	✓	96.7%	91.2%

[†] Phi-4 14B NM rates use trimmed curves (excluding last column; see Appendix V). Phi-4-reasoning NM computed on first 16,384 generated tokens ($n=180$); truncation artifact $\Delta \leq 3$ pp (§3.1).

Table 47: Exit accuracy by layer ($n=200$). 1.5B: R1-Distill-Qwen-1.5B vs. Qwen2.5-1.5B-Instruct (28 layers). 14B: Phi-4 base (40 layers).

Model	L8	L12	L16	L20	L24	L28	L32	L36
R1-Distill-1.5B	5.4	7.7	10.9	41.6	81.3	—	—	—
Qwen2.5-1.5B	2.2	3.3	5.3	32.5	79.2	—	—	—
Phi-4 14B base	18.6	19.7	19.9	24.0	35.5	55.7	70.5	91.2

Answer magnitude. Stratifying by the numerical value of the final answer reveals a significant positive correlation with non-monotonicity ($\rho=0.35$, $p=1.5 \times 10^{-15}$; partial $\rho=0.18$ after controlling for trace length). Questions with large answers ($>1,000$) exhibit 15.6% non-monotonicity versus 9.9% for small answers (<10), indicating that large-number tokens have inherently less stable intermediate-layer representations.

X End-to-End Exit Details

Table 53 reports fixed-layer exit accuracy for all four models in the 2×2 design (§5.1). Question-level accuracy requires every token to match the final-layer prediction; all models yield 0% at every sub-final layer.

Alternative cosine-similarity definition. On a smaller pilot ($n=50$ questions), final-layer cosine similarity achieves 20% Q_{Acc} at $\tau=0.9$, but with negligible savings (0.03%), confirming that no threshold yields a useful accuracy-savings trade-off.

Probe exit accuracy details. Table 54 reports logistic regression probe AUC on PCA-reduced hidden states (§5.1). AUC decreases with depth (0.87 at L12 $\rightarrow$ 0.78 at L28), indicating that at early layers correct-exit tokens are easy outliers but fewer than 2% have matured.

Y Logistic Regression Validation of Token-Role Taxonomy

In R1-Qwen traces (1,114 questions), token proportions are: numerical 18.4%, special 23.5%, reasoning keyword 4.6%, other 53.6%.

To test whether the four-category token role taxonomy has predictive validity beyond descriptive statistics, we fit a logistic regression predicting non-monotonic convergence (binary: NM vs. monotonic, $\theta=0.03$) from token role (categorical: numerical, special, reasoning keyword, other as reference). Table 55 reports odds ratios (OR) for all three models. R1-Llama uses the on-server replication cache (113k tokens, 100q, $\theta=0.03$); the original full-data cache (199k tokens, MATH-500 100q) resided on now-retired hardware and yielded pseudo- $R^2=0.018$ ($1,179 \times$ random; §4.3). The replication cache shows a similar qualitative direction but smaller magnitude (pseudo- $R^2=0.010$, $409 \times$ random), consistent with sample-size sensitivity. Both caches exceed the random four-way partition by two-to-three orders of magnitude.

³Phi-4 logistic regression uses 200 questions (180 from the main profiling plus 20 supplementary); Table 2 reports the 180-question subset (2.8M tokens).

Table 48: V2a gate on QwQ-32B (MATH-500 $n=500$, 5-fold nested CV). The gate recovers 61.0% Q_{Acc} at $\gamma=0.99$ despite standard CALM collapsing to 2.4%.

γ	Q_{Acc} (%)	Savings (%)
0.99	61.0 ± 2.4	4.98 ± 0.44
0.95	12.0 ± 3.2	12.29 ± 0.50
0.90	1.8 ± 1.9	18.23 ± 0.32
0.50	0.0 ± 0.0	56.10 ± 0.79

Table 49: V2a gate on Phi-4-reasoning-14B (cross-architecture transfer from R1-Qwen, GSM8K $n=500$, nested CV). Gate signal transfers (48.0% Q_{Acc} at $\gamma=0.99$; 12.70% savings), though Q_{Acc} is lower than R1-Qwen (63.0%).

γ	Q_{Acc} (%)	95% CI	Savings (%)
0.99	48.0 ± 7.3	[39.0, 57.0]	12.70 ± 0.71
0.95	28.2 ± 2.3	—	17.09
0.90	7.8 ± 3.6	—	23.94
0.50	0.0 ± 0.0	—	55.20

Token role is a highly significant correlate in all three models (likelihood ratio test $p \approx 0$). The effect direction reverses across models: numerical tokens are the strongest NM correlate in R1-Qwen (OR=6.76) but the weakest in Phi-4 (OR=0.20), while special tokens show the opposite pattern. R1-Llama shows intermediate effect sizes with the same qualitative direction as Phi-4 (numerical OR=0.50, special OR=1.12). Reasoning keywords are consistently the most monotonic role (OR=0.12–0.59), aligning with the representation immaturity pattern identified in §4.3. The modest pseudo- R^2 (0.01–0.10) indicates that token role captures significant but not dominant variance—consistent with position, context, and question difficulty as additional contributing factors. A permutation baseline confirms this signal is genuine: random 4-class partitions (preserving class proportions, 1,000 permutations) yield pseudo- $R^2 \approx 10^{-5}$, making the actual taxonomy $>1,000\times$ more predictive ($p < 0.001$ for both R1-Qwen and R1-Llama). The absolute magnitude remains modest, supporting our framing of token roles as a diagnostic lens rather than a dominant explanatory variable.

Exit correctness prediction. Token role also significantly predicts per-layer exit correctness (top-1 match with final layer; R1-Qwen, 1.49M tokens, $p \approx 0$ at all layers). Table 56 reports odds ratios at three representative layers. All non-“other” roles show OR < 1 —they are *less* likely to exit correctly. The effect is strongest at early layers (reasoning keywords OR=0.08 at L8) and attenuates with depth (OR=0.91 at L28), consistent with the representation immaturity pattern: specialized tokens require more layers to reach correct predictions. L32 per-token exit correctness data is not available in our cache; results are reported through L28. Combined with the NM analysis above, token role is a significant correlate of *both* convergence stability (H1) and exit accuracy (H2), providing functional validation across both dimensions of the dual failure pattern.

Z GSM8K Cross-Benchmark Validation

To assess whether our findings generalize beyond MATH-500, we replicate the end-to-end early exit analysis on GSM8K ($n=50$ questions, 8,289 tokens) using the identical 7-layer exit grid [8, 12, 16, 20, 24, 28, 32] and R1-Qwen3-8B. The small sample size yields wide binomial confidence intervals ($\approx \pm 12.7$ pp on question-level accuracy); we therefore emphasize directional consistency rather than point-estimate comparisons.

The core pattern replicates: tuned lens projection dramatically expands oracle savings (+41.4 pp on GSM8K vs. +46.7 pp on MATH-500), while CALM deployment quality remains constrained by compound error. Confidence-based CALM at $\tau=0.99$ achieves 30.0% Q_{Acc} on GSM8K (vs. 18% on MATH-500); however, this difference falls within the binomial CI and should not be interpreted as a statistically meaningful improvement.

Compound Error vs. Trace Length — R1-Qwen3-8B

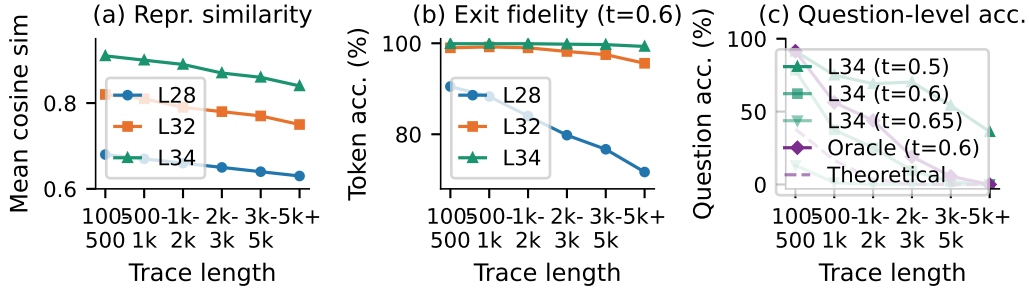


Figure 8: Compound error vs. trace length. (a) Per-token cosine similarity declines modestly with length. (b) Token-level exit fidelity ($\cos \geq 0.60$) is roughly stable at $\sim 99.5\%$. (c) Question accuracy (all tokens correct) decays sharply, matching the p^T compound error model. Data: 500 GSM8K questions, DeepSeek-R1-Qwen3-8B.

Table 50: Compound error under independence: $Q_{\text{Acc}} = p^T$. Theoretical values for representative per-token accuracies and trace lengths typical of reasoning models. Bottom rows: observed operating points (R1-Qwen3-8B, GSM8K) and required p for 50% question accuracy.

Per-token acc. (p)	$T=500$	$T=1,000$	$T=3,000$
0.990	0.66%	$<0.01\%$	$\approx 0\%$
0.995	8.1%	0.66%	$\approx 0\%$
0.999	60.6%	36.8%	5.0%
0.9995	77.9%	60.6%	22.3%
0.9999	95.1%	90.5%	74.1%
<i>Observed operating points (R1-Qwen3-8B, avg $T \approx 3,000$)</i>			
CALM $\cos \tau=0.95$: $p=0.947$	$\approx 0\%$	$\approx 0\%$	$\approx 0\%$
CALM $\cos \tau=0.97$: $p=0.992$	1.8%	0.03%	$\approx 0\%$
Tuned lens $\tau=0.99$: $p=0.990$	0.66%	$<0.01\%$	29.4% [†]
<i>Required p for $Q_{\text{Acc}} \geq 50\%$</i>			
$p_{\min}$	99.861%	99.931%	99.977%

[†] Observed Q_{Acc} exceeds p^T prediction: tuned projection corrects correlated errors.

Token type distribution. GSM8K traces contain fewer LaTeX symbols and explicit reasoning keywords than MATH-500: special tokens 13.4% (vs. 23.5%), reasoning keywords 1.9% (vs. 4.6%), numerical 20.2% (vs. 11.0%), other 64.5% (vs. 60.9%). Aggregate patterns—the tuned lens oracle gap and compound error ceiling—generalize across benchmarks, but per-type proportions are benchmark-dependent (GSM8K contains fewer LaTeX symbols and explicit reasoning keywords).

Z.1 Cross-Benchmark V2a Transfer (GSM8K)

To test whether V2a exit decisions generalize beyond the training benchmark, we evaluate the MATH-500-trained gate on GSM8K (1,319 questions) using 5-fold stratified CV with a threshold sweep.

The higher Q_{Acc} on GSM8K (86.4% vs. 63.0%) likely reflects GSM8K’s shorter reasoning traces (median ~ 85 tokens vs. MATH-500’s $\sim 3,000$), which reduce compound-error accumulation. Conversely, the conservative exit behaviour at $\gamma=0.99$ yields only 2.3% savings, confirming that threshold calibration is benchmark-specific. V2a exit *decisions* thus transfer cross-benchmark, but *efficiency* requires per-task calibration.

Table 51: Non-monotonicity rate by trace position ($n=1,466$ questions, $\theta=0.03$). Per-type breakdown shows reasoning keywords exhibit the strongest temporal decay ($3.2\times$).

Segment	Overall	Numerical	Special	Reasoning KW	Other
Q1 (0–25%)	17.8%	39.6%	21.2%	4.6%	10.6%
Q2 (25–50%)	16.4%	37.0%	19.5%	2.3%	9.1%
Q3 (50–75%)	15.3%	34.9%	18.4%	1.8%	8.4%
Q4 (75–100%)	18.7%	38.9%	23.8%	1.4%	10.3%

Table 52: Trace-length stratification ($n=500$, R1-Qwen3-8B, 5 quintiles of 100). Non-mono rate is stable; Q_{Acc} decay is purely compound error.

Quintile	Token range	Non-mono (%)	CosSim L34	Q_{Acc} ($\theta=0.60$)
Q1 (short)	274–967	11.4 ± 4.6	0.903	47.0%
Q2	967–1,695	13.7 ± 5.7	0.889	26.0%
Q3	1,695–2,839	12.7 ± 6.7	0.871	14.0%
Q4	2,839–4,549	11.1 ± 5.9	0.859	2.0%
Q5 (long)	4,549–16,384	10.6 ± 5.0	0.845	0.0%

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: The abstract and §1 state two specific contributions: (1) dual failure pattern (oscillatory dynamics distillation-specific, representation immaturity reasoning-general), separable via thinking mode ablation on one architecture, and (2) projection bottleneck and learned gating—tuned lens analysis revealing projection quality as the primary in-distribution bottleneck (+45.6pp) while compound error remains the binding deployment constraint; feature importance analysis confirms tuned lens confidence features as the primary gating drivers. Scale boundaries (7B–32B confirmed, 1.5B reversal) are reported in §7. Each claim is supported by corresponding sections (§3–6, Appendix U) with quantitative evidence and confidence intervals.

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Table 53: Fixed-layer exit (raw logit lens): question-level accuracy is 0% at every layer for all models. Token accuracy (question-level mean) at deepest sub-final layer.

Model	Deepest Exit	Rel. Depth	Token Acc	Question Acc	Avg Tokens/Q
R1-Qwen3-8B	L32	89%	62%	0%	~2,980
R1-Llama	L28	88%	66%	0%	~620
Qwen3-8B	L32	89%	73%	0%	~3,847
Llama-3.1-8B	L28	88%	73%	0%	~295

	L12	L16	L20	L24	L28
Exit accuracy (%)	1.3	1.5	3.7	17.2	32.0
Probe AUC	0.87	0.86	0.84	0.78	0.78

Table 54: Logistic regression probe AUC on PCA-reduced hidden states (128 dim, 400/100 question-level split). Per-type breakdown in Appendix H.

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4. Experimental Result Reproducibility

Table 55: Logistic regression: token role predicting non-monotonic convergence. OR > 1 indicates higher NM probability relative to the “other” reference category. All coefficients $p < 10^{-200}$. R1-Llama uses the on-server replication cache (113k tokens, $\theta=0.03$); the original full-data cache (199k tokens) yielded pseudo- $R^2=0.018$, $1,179\times$ random (§4.3).

Role	R1-Qwen (1.49M)		R1-Llama (0.11M)		Phi-4-reas. (3.12M) ³	
	OR [95% CI]	NM	OR [95% CI]	NM	OR [95% CI]	NM
other (ref)	—	8.1%	—	22.2%	—	22.5%
numerical	6.76 [6.69, 6.84]	37.3%	0.50 [0.47, 0.53]	12.5%	0.20 [0.19, 0.20]	5.4%
special	1.88 [1.86, 1.91]	14.2%	1.12 [1.09, 1.16]	24.2%	2.67 [2.65, 2.68]	43.6%
reas. keyword	0.59 [0.57, 0.60]	4.9%	0.54 [0.49, 0.60]	13.4%	0.12 [0.11, 0.12]	3.3%

LR test: R1-Qwen $\chi^2=116,326$; R1-Llama $\chi^2=1,126$; Phi-4 $\chi^2=176,538$ (all $p\approx 0$)
McFadden pseudo- R^2 [McFadden, 1974]: R1-Qwen = 0.098; R1-Llama = 0.010; Phi-4 = 0.050

Table 56: Logistic regression: token role predicting exit correctness at three layers (R1-Qwen, 1.49M tokens). OR < 1 indicates lower exit accuracy relative to “other.” All coefficients $p\approx 0$. All 95% CIs narrower than ± 0.02 given $N=1.49M$; omitted for brevity.

Role	L16 (OR)	L24 (OR)	L28 (OR)
other (ref)	—	—	—
numerical	0.29	0.99	0.78
special	0.26	0.72	0.91
reas. keyword	0.17	0.79	0.91

LR test χ^2 : 6,690 (L16), 3,393 (L24), 2,445 (L28)
McFadden pseudo- R^2 : 0.029 (L16), 0.002 (L24), 0.001 (L28)

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Table 57: Cross-benchmark comparison of early exit metrics on the same 7-layer grid [8, 12, 16, 20, 24, 28, 32]. Question-level accuracy (Q_{Acc}) is reported with sample size; oracle savings aggregate over all tokens.

Metric	GSM8K ($n=50$)	MATH-500 ($n=100$)
Raw oracle savings	11.4%	10.4%
Tuned oracle savings	52.8%	57.1%
Oracle Δ (tuned – raw)	+41.4 pp	+46.7 pp
Tuned coverage	89.7%	93.3%
CALM conf Q_{Acc} ($\tau \leq 0.95$)	0% [†]	7.2% ^{††}
CALM conf Q_{Acc} ($\tau = 0.99$)	30.0% (13.2% savings)	18%
CALM cosine Q_{Acc} ($\tau = 0.99$)	86.0% (0.45% savings) [‡]	—

[†]At $\tau \leq 0.95$, 100% of GSM8K tokens exit early, yielding 0% Q_{Acc} . Both values reflect the failure of confidence-based early exit at aggressive thresholds.

^{††}MATH-500 CALM Q_{Acc} from 9-layer main experiment (Table 4); oracle and coverage rows use the 7-layer grid for direct comparison.

[‡]Cosine CALM at $\tau = 0.99$ exits only 0.88% of tokens (savings=0.45%), effectively a no-exit baseline. At $\tau \leq 0.95$, all tokens exit and $Q_{\text{Acc}} = 0\%$.

Table 58: V2a gate transfer from MATH-500 to GSM8K. The gate’s exit *accuracy* generalises strongly (86.4% Q_{Acc} at $\gamma = 0.99$ vs. 63.0% on MATH-500), while savings drop (2.3% vs. 6.28%), indicating that per-task threshold calibration is needed for efficiency.

γ	Q_{Acc} (%)	Savings (%)	MATH-500 reference	
			Q_{Acc} (%)	Savings (%)
0.50	0.0 ± 0.0	37.7 ± 0.6	—	—
0.90	15.1 ± 4.1	12.4 ± 0.6	—	—
0.95	40.5 ± 3.4	7.7 ± 0.4	—	—
0.99	86.4 ± 1.6	2.3 ± 0.2	63.0 ± 5.2	6.28

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